

Indian Knowledge System

Last-Night Exam Study Guide · Units 1–5

Built from your Unit notes, Model Question Paper, SEE Feb 2026, CIE 1/2/3 solutions, question banks and the prescribed textbook (Mahadevan, Bhat & Pavana, PHI).

How to read this guide

Blue highlight	Must-remember fact / keyword. Examiners tick these.
IMPORTANT blue box	Asked (or very likely to be asked) in past papers: SEE Feb 2026, Model QP, CIE 1–3, QBs.
BOOK p.165 ↗	Clickable link: opens the textbook PDF at that section. Use it for anything you want to read in full. Links point to F:\th_year_resources_2022_scheme_RVCE\Indian Knowledge Systems\introduction-to-indian-knowledge-system-TB.pdf , so they work only on this laptop with that file in place (open the guide in Chrome/Edge or Acrobat). Printed page number is shown.
Orange box	Conflict between your sources, or a trap. Read these.

Exam pattern (from MQP and SEE Feb 2026)

- **Part A:** 60 MCQs × 1 mark = 60 on OMR, about 12 per unit (mostly one-liners).
- **Part B:** 5 questions × 8 marks (a = 4 + b = 4) = 40. One question per unit, each with an **either/or** choice. So **one 8-mark answer from each unit**: learn two "either" options per unit well.
- Total 100 marks (SEE paper split into 60 + 40, 1.5 h each).

Priority order for tonight: (1) Rapid-fire facts (last section) · (2) Part B answer outlines (second-last section) · (3) Unit 2 (Pāṇini, zero, Piṅgala, Pramāṇa) and Unit 3 (Pañchāṅga, Wootz, Delhi Pillar, perfumes) — these have the most testable detail · (4) Units 1, 4, 5 by headings.

Tip: the **SEE Feb 2026 paper** (1.5 h; Part A 60 × 1 mark MCQs, Part B 40 marks) asked in Part B: Vedas vs Vedāṅgas (4 differences); historicity of IKS; Pāṇini "father of linguistics"; decimal place-value; Anumāna / Pañca-avayava; Gaṇas & de Bruijn; 4 mathematicians; perfume making; Pañchāṅga; Delhi Iron Pillar; Pitta doṣa; Nāgara vs Drāviḍa; 64 Kalās in modern education; Vāstu in modern design; advisors in Rāma's administration; TKDL; Arthaśāstra sources of state income. The new paper will likely rotate neighbouring topics, so all are covered below.

UNIT 1 · Introduction to Indian Knowledge Systems

1.1 What is IKS? (Perception & prologue)

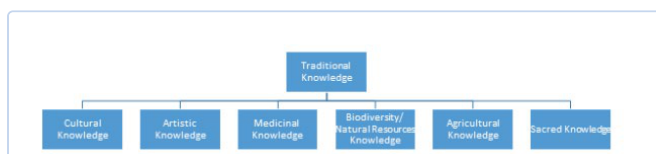
- IKS = body of knowledge and practices developed in India over **thousands of years**: philosophy, mathematics, medicine (Ayurveda), arts, architecture, agriculture, understanding of nature. Passed down through generations; promotes **harmony with nature and society**.
- Colonial education marginalised indigenous knowledge; Western epistemology dominated, creating a gap between learners and their roots. **NEP 2020** advocates reviving IKS through curriculum integration at all levels (interdisciplinary learning, indigenous languages, traditional wisdom for sustainability).
- Modern synthesis examples: traditional farming + modern tools; Ayurveda + modern healthcare.
- Languages: Sanskrit carries much of the knowledge; the motto "*Satyam vada, dharmaṃ cara*" (Taittirīya Upaniṣad; SEE MCQ). "*Sā vidyā yā vimuktaye*" = knowledge that liberates.

BOOK p.3 ↗

BOOK p.6 Importance ↗

1.2 Indian Traditional Knowledge (TK): nature, scope, kinds

- TK / indigenous knowledge = knowledge, know-how, skills, practices **developed, maintained and passed from generation to generation**, often central to a community's cultural/spiritual identity.
- **Scope**: agriculture, science, technology, ecology, medicine (Ayurveda, Yoga, Unani, Siddha, Naturopathy), biodiversity, plus cultural expressions (signs, symbols).
- **How it is expressed**: mostly **oral**: tales, songs, folklore, proverbs, rituals, dance, paintings, carvings; belongs to the community as a group.
- **Kinds of TK (figure)**: Cultural, Artistic, Medicinal, Biodiversity/Natural-resources, Agricultural, Sacred knowledge. (SEE MCQ: "ecological, technical, social".)



Kinds of Traditional Knowledge (your notes)

IMPORTANT TK vs Modern science (MQP Q1b, CIE-1)

- TK: **holistic, experiential, community-based, oral, sustainable**; knowledge = path to truth (*satya*) and harmony of self–nature–universe.
- Modern/Western: **analytical, reductionist, empirical, documented**; observation + experiment to explain/control nature.

Threats to TK & why protect it

Threats	Need to protect
• Language threat: indigenous languages (e.g. Sanskrit for Ayurveda) decline • External pressures: migration, environment, modern lifestyle, globalisation • Biopiracy: commercial exploitation of natural biochemicals/genetic material (turmeric patent by an MNC) • Colonisation, capitalism/MNCs, consumerism	• Preserve culture & identity • Protect livelihoods • Health benefits (traditional medicine for large populations) • Ecological benefits: e.g. sacred groves conserve biodiversity

1.3 Transition from traditional to Western knowledge **IMPORTANT**

- **Traditional**: Gurukula system; Nalanda, Takshashila; holistic (philosophy + science + arts + ethics); knowledge = *Vidya*, sacred, aimed at **self-realisation and societal harmony**; oral + Sanskrit texts.
- **British rule**: English medium, new curricula, modern institutions; focus shifted to **empirical, utilitarian, job-oriented** learning. Created a divide (an *epistemological shift*; MQP Q2 answer B).

- Silver lining: documentation and global exposure of Indian heritage; today we seek **synthesis** of IKS with modern science.

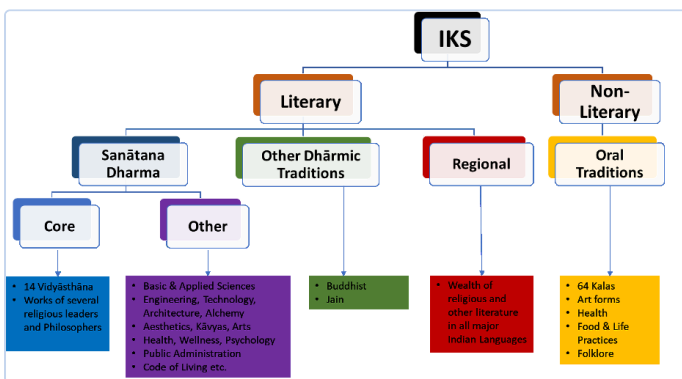
1.4 Historicity of IKS **IMPORTANT** (SEE Feb 2026 1b)

Strand	Key facts
Vedic period (c. 1500–500 BCE)	Origin; four Vedas (cosmology, spirituality, ethics); Upanishads: self (Ātman) and ultimate reality (Brahman).
Epics & classics	Rāmāyaṇa, Mahābhārata (moral guidance); Bhagavad Gītā (in Mahābhārata); Kālidāsa's <i>Śakuntalā</i> ; Patañjali's <i>Yoga Sūtras</i> (holistic well-being).
Philosophy	Six schools: Nyāya, Vaiśeṣika, Sāṃkhya, Yoga, Mīmāṃsā, Vedānta.
Mathematics	Āryabhaṭa, Brahmagupta, Bhāskarācārya: zero, decimal system, trigonometry; Brahmagupta: quadratic equations.
Astronomy	Āryabhaṭa: Earth's rotation, circumference; Varāhamihira: <i>Bṛhat Saṃhitā</i> (astronomy + meteorology).
Medicine	Ayurveda: Charaka Saṃhitā (internal medicine), Suśruta Saṃhitā (surgery).
Learning centres	Takshashila (c. 600 BCE; medicine, law, military science). Nalanda (c. 5th c. CE; library <i>Dharmagañja</i> ; students from China, Korea, Central Asia).
Manuscripts & scripts	Palm leaf, birch bark; Brāhmī, Kharoṣṭhī, later Devanāgarī. Architecture texts: Mānasāra, Mayamata, Samarāṅgaṇa-sūtradhāra.

BOOK p.16 Historicity ↗

BOOK p.19 Oral tradition & sūtras ↗

1.5 The IKS Corpus: classification framework **IMPORTANT**



IKS corpus tree (your notes). Memorise this shape.

- **Literary** = **Sanātana Dharma corpus** (Core: Vedas + Vedāṅgas, 14 Vidyāsthānas; Other: basic & applied sciences, engineering, architecture, alchemy, health, psychology, public administration, code of living) + **Other Dhārmik traditions** (Buddhist, Jain) + Regional literature.
- **Non-literary** = oral traditions: 64 Kalās, art forms, health practices, food, folklore.
- Buddhist: Nāgārjuna's *Rasaratnākara* (1st c. CE, alchemy). Jain: mathematics as *Gaṇitānuyoga*, *Tattvārtha Sūtra*.
- Nītiśāstra: **Rāja-nīti** (governance) and **Sāmānya-nīti** (ethics).
- CIE-1 version: Vedic corpus (Śruti) · Post-Vedic (Smṛti: Itihāsa, Purāṇas, Dharmaśāstras) · Traditional sciences · Folk & tribal knowledge.

BOOK p.9 Corpus ↗

BOOK p.13 Caturdaśa-Vidyāsthāna ↗

1.6 The Vedic Corpus **IMPORTANT**

- **Veda** = "knowledge", from root *vid* (to know). Śruti = "that which was heard": **oral tradition / revelation** (MQP Q1: B). Written form c. 1600 BCE; no definite date as oral transmission was precise.
- Four Vedas = **Chaturveda**: Ṛg, Yajur, Sāma, Atharva.

Veda	Meaning / nature	Facts to remember
R̥gveda	"Praise" · hymns; oldest	1,028 hymns, 10,552 verses, 10 Maṇḍalas ; sūktas: Puruṣa (10.90), Nāsadiya (10.129), Hiraṇyagarbha (10.121); 5 of 21 śākhās survive (Śākala, Bāṣkala, Āśvalāyana, Śāṅkhāyana, Māṇḍūkāyana). Aitareya Upaniṣad: Mahāvākya <i>Prajñānam Brahma</i> .
Yajurveda	Prose mantras · "book of rituals"	Two branches: Śukla (mantra and brāhmaṇa separate, "bright") and Kṛṣṇa (mixed, "dark") . 101 śākhās once, 6 now (4 Kṛṣṇa: Taittirīya, Maitrāyaṇī, Kaṭha, Kapiṣṭhala; 2 Śukla: Kāṇva, Mādhyandina).
Sāmaveda	Songs / melodies · "book of songs"	Shortest: 1,549 verses ; root of Indian classical music and dance; 3 recensions survive (Kauthuma, Jaiminīya, Rāṇāyaṇīya); Chāndogya and Kena Upaniṣads.
Atharvaveda	Formulas/spells for diseases and daily life	730 hymns, ~6,000 mantras, 20 books ; Muṇḍaka, Māṇḍūkya, Praśna Upaniṣads; medical/surgical verses (hymn 4.15, fracture).

IMPORTANT Four divisions of each Veda (SEE Feb 2026 2b, MQP 2a)

1. **Saṃhitā**: oldest layer: mantras, hymns, prayers, benedictions.
2. **Brāhmaṇa**: prose commentary on hymns and rituals; how to perform yajña.
3. **Āraṇyaka**: "forest texts": philosophy/symbolism behind the ritual (for hermits).
4. **Upaniṣad**: teacher–student dialogues on the self (Ātman) and Brahman; **Vedānta = end of the Vedas** (CIE-1 MCQ).

[BOOK p.28 Vedas ↗](#)

[BOOK p.30 Four Vedas ↗](#)

[BOOK p.33 Four divisions ↗](#)

1.7 Vedāṅgas: the six limbs **IMPORTANT**

Vedāṅga	Subject	Body part
Śikṣā	Phonetics, pronunciation, accent	Nose
Vyākaraṇa	Grammar	Mouth / face
Nirukta	Etymology, meaning of words	Ears
Chandas	Metre / prosody	Feet
Kalpa	Ritual procedure	Hands
Jyotiṣa	Astronomy / timekeeping	Eyes

- **Śikṣā**: Śikṣāvalli chapters: varṇa, svara, mātṛā, bala, sāma, santāna.
- **Kalpa sūtras (4)**: Śrauta (sacrifices), Gṛhya (domestic), Dharma (law), **Śulba** (measurement of fire altars; source of geometry).
- **Vyākaraṇa**: Pāṇini c. 400 BCE; Maheśvara Sūtras (14, from Naṭarāja's ḍamaru); Patañjali's *Mahābhāṣya*.
- **Nirukta**: Yāska's *Nighaṇṭu* (c. 600 BCE); "all names come from verbs".
- **Chandas**: Gāyatrī = 3 pādas × 8 = 24 syllables; Anuṣṭup = 4 × 8 = 32 (classical śloka).
- **Jyotiṣa**: Varāhamihira's *Pañcasiddhāntikā* (5 siddhāntas); Pañchāṅga.

IMPORTANT Vedas vs Vedāṅgas: four differences (SEE Feb 2026 1a)

	Vedas	Vedāṅgas
Purpose	Holy texts: origin of world, reason for existence	Ancillary studies to preserve, interpret, perform the Vedas
Structure	4 collections	6 limbs
Content	Hymns, incantations, rituals	Metre, grammar, linguistic analysis, astronomy
Origin	Among the oldest scriptures, c. 1500–1200 BCE	Last treatises of Vedic literature

[BOOK p.38 Vedāṅgas ↗](#)

[BOOK p.39 Śikṣā ↗](#)

[BOOK p.43 Kalpa ↗](#)

[BOOK p.45 Jyotiṣa ↗](#)

1.8 Indian philosophical systems (Darśanas) **IMPORTANT**

Origins in R̥gveda hymns (Nāsadiya, Puruṣa, Vāk) → flowering in Upaniṣads. **Āstika** = accepts Vedas as valid (6 schools). **Nāstika** = denies Vedas (Cārvāka, Bauddha, Jaina).

School	Founder / text	Core idea
Nyāya	Akṣapāda Gautama · Nyāya Sūtras	Logic & epistemology ; 4 pramāṇas (perception, inference, comparison, testimony).
Vaiśeṣika	Kaṇāda · Vaiśeṣika Sūtra (6th c. BCE)	Atomism, pluralism ; physical reality made of atoms; originally 2 pramāṇas (perception, inference).
Sāṃkhya	Kapila · Sāṃkhya Kārikā	Oldest orthodox; dualism of Puruṣa (consciousness) and Prakṛti (matter) ; 3 guṇas (sattva, rajas, tamas); 25 elements.
Yoga	Patañjali · Yoga Sūtras	Sāṃkhya + God; 8 limbs (aṣṭāṅga) ; goal kaivalya.
Pūrva Mīmāṃsā	Jaimini	Interpret Vedas; ritual action (karma-kāṇḍa); salvation through Vedic action.
Vedānta (Uttara Mīmāṃsā)	Bādarāyaṇa · Brahma Sūtras	Upaniṣadic knowledge (jñāna-kāṇḍa). Advaita (Ātman = Brahman), Viśiṣṭādvaita (attributes), Dvaita (3 realities).
Cārvāka (Lokāyata)	Brhaspati	Only pratyakṣa valid; "extreme nāstika"; material.
Bauddha	Gautama Buddha	Non-theistic; Four Noble Truths, Noble Eightfold Path to end <i>dukkha</i> .
Jaina	24 Tīrthaṅkaras; Mahāvīra (6th c. BCE)	Anekāntavāda (many viewpoints), ahiṃsā ; only Kevalīs know full truth.

[BOOK p.56 Overview ↗](#)
[BOOK p.59 Sāṃkhya & Yoga ↗](#)
[BOOK p.63 Nyāya/Vaiśeṣika ↗](#)
[BOOK p.65 Mīmāṃsā/Vedānta ↗](#)
[BOOK p.70 Jaina ↗](#)
[BOOK p.72 Bauddha ↗](#)
[BOOK p.75 Cārvāka ↗](#)

UNIT 2 · Foundations of Maths, Linguistics & Knowledge Framework

2.1 Pāṇini & the Aṣṭādhyāyī (linguistics) **IMPORTANT**

- **Pāṇini**: born in Gandhāra; c. 5th–6th c. BCE (some sources c. 400 BCE). **Father of linguistics**. His morphological analysis was ahead of any Western theory until the 20th century.
- **Aṣṭādhyāyī** = "eight chapters" (*aṣṭa* = 8, *adhyāya* = chapter): **3,983 sūtras**; 8 chapters × 4 quarters (*pādas*) = 32 sections. MQP Q14: Eight.
- Sūtra style = concise rules, memorised orally. Unique **metalanguage**; structure compared to **Backus–Naur Form (BNF)**.
- **Tri-Muni** of Sanskrit grammar: Pāṇini (sūtras) · Kātyāyana (vārtikas, 4th c. BCE) · Patañjali (*Mahābhāṣya*, 2nd c. BCE).
- Four parts of Pāṇinian grammar: **Śivasūtra/Maheśvara Sūtras** (14 phoneme sūtras; phonology; *pratyāhāra* shorthand) · Aṣṭādhyāyī (rules) · **Dhātupāṭha** (list of verbal roots, ~2,200) · **Gaṇapāṭha** (classes of noun stems).

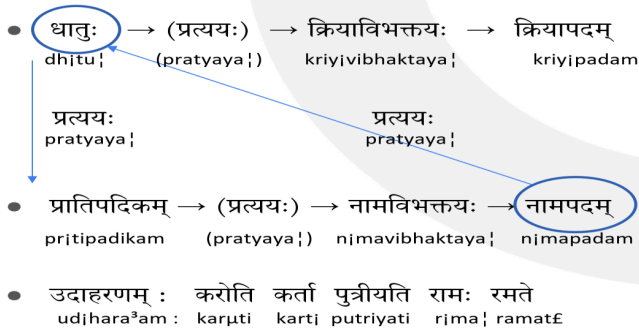


Pāṇini on the ₹500 stamp

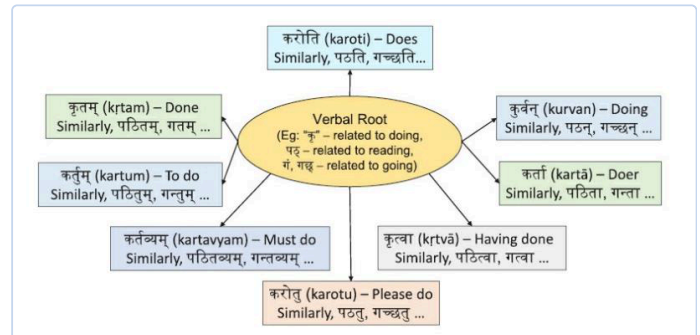
Word generation: why it is "computational"

- **Root + suffix + rules → word**. Verbal root (*dhātu*) + suffix (*pratyaya*) → verb form; nominal stem (*prātipadika*) + **sup** suffix → noun form. Verb suffix = **tiṅ**. (SEE MCQs: noun declension suffixes = **sUP**; basic generator = **dhātu / prātipadika**.)
- Noun: 7 cases × 3 numbers = **21** sup suffixes. Verb: 3 persons × 3 numbers = **9** tiṅ suffixes.
- Example root *kṛ* (to do; in Dhātupāṭha "ḍukṛñ"): *karoti*, *kurvan*, *kartā*, *kṛtvā*, *karotu*, *kartavyam*, *kartum*, *kṛtam*. The **same pattern applies to ~2,200 verbs** (*paṭh* → *paṭhati*..., *gam* → *gacchati*...).
- Derivation is step-by-step and recursive: apply a rule if its conditions hold; repeat till no rule applies. **Vocabulary is not fixed: new valid words can be generated** → Sanskrit stays relevant; suited to NLP/AI.

The Word Generation Scheme



Word generation scheme



Derivatives of root *kṛ*

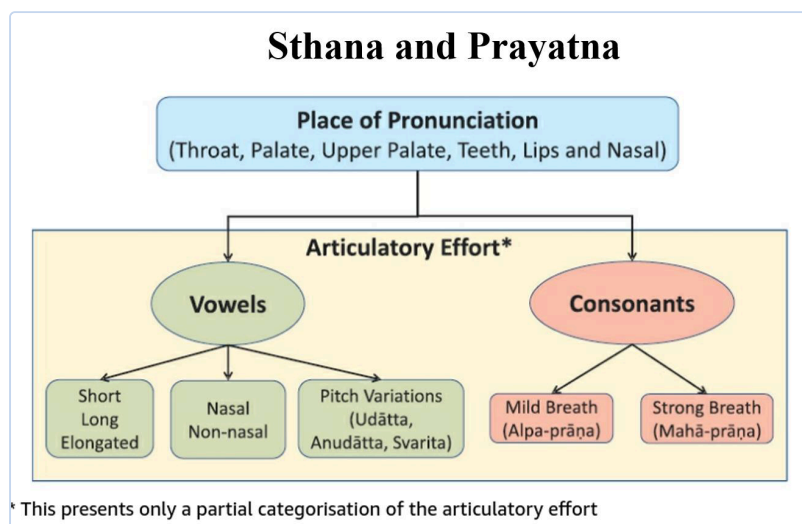
Pāṇini's grammar	Computer science (CFG)
Sūtra (rule)	Production rule
Abstract categories (<i>prātipadika</i> , <i>dhātu</i> , <i>sup</i> , <i>tiṅ</i>)	Non-terminals (NP, VP)
Sounds / morphemes	Terminals
Sequential + recursive rule application (<i>anuvṛtti</i>)	Recursive expansion
Paribhāṣā sūtras (rules about rules)	Meta-grammar
Generative and transformative	Only generative

Example: *Rāmaḥ phalam khādati* = *S* → NP VP; NP → N SUFFIX; VP → V NP. Compound (*samāsa*): *tatpuruṣaḥ samāsaḥ*, *rāja-putra* = COMPOUND → N N.

IMPORTANT Why "father of linguistics"? Traditional: preserved Vedic Sanskrit accurately; 3,983 rule system; standard, universally accepted model. Modern: formal generative grammar; algorithmic, recursive, rule-based (BNF-like); influences computational linguistics, NLP, AI; oldest and most complete descriptive model of language.

Phonetics (Śikṣā / sound science)

- Linguistics = sound (listening/speaking) + script (reading/writing). Language-related Vedāṅgas: **Śikṣā** (phonetics), **Nirukta** (etymology), **Vyākaraṇa** (grammar).
- Sthāna** (MQP/SEE MCQ) = **place of articulation** (throat, palate, upper palate, teeth, lips, nasal); **Prayatna** = articulatory effort (e.g. mild breath/*alpaprāṇa* vs strong breath/*mahāprāṇa*; vowels short/long/nasal).



Sthāna and Prayatna

BOOK p.116 Pāṇini ↗

BOOK p.117 Phonetics ↗

BOOK p.120 Patterns ↗

BOOK p.122 Computational concepts ↗

BOOK p.128 Sentence logic ↗

2.2 Number system, zero and decimal place-value **IMPORTANT**

- Decimal place-value system** (10 symbols 0–9; position decides value) is India's greatest gift. Evidence: Ashokan edicts (3rd c. BCE); Brāhmī numerals; legal document **594 CE** and inscription **876 CE** (Gwalior) with place value and zero; Al-Bīrūnī's notes; Ifrah's research.
- Roman numerals are impractical: 432,000 needs M written 432 times.
- Zero (śūnya)**: first formal definition word "śūnya" in Piṅgala's *Chandaḥśāstra*. **Brahmagupta (628 CE, Brahmasphuṭasiddhānta)**: zero as a **number** with rules ($a+0=a$, $a \times 0=0$; $a/0$ undefined; debts/fortunes). Āryabhaṭa used place value. Bakhshālī manuscript: dot for zero (carbon-dated 3rd–4th c. CE). Gwalior temple inscription (9th c.) = earliest recorded zero in stone.
- Transmission**: India → Arabs (Kaṅka translated Sanskrit; **Al-Khwārizmī**, Al-Samaw'al) → Spain/Europe (Fibonacci's *Liber Abaci*, 1202). Words *algorithm*, *algebra*.
- Large numbers**: Śukla Yajurveda 17.2 names powers of ten to 10^{12} (eka, daśa, śata, sahasra, ayuta, niyuta, prayuta, arbuda, nyarbuda, samudra..., parārdha). Pattern = *daśaguṇottara saṁjñā*. Lalitavistara: Buddha counts to *tallakṣaṇa* (10^{53}) and beyond (up to 10^{421}).
- Measurement** (QB): three generic measures = **length, weight, time**. **Aṅgula** = basic Vedic unit (Śulba). Arthaśāstra: Dhanus = 96 aṅgula; Gṛhapatya-dhanus = 108 aṅgula (roads). Indus cubit ≈ 51.8 cm; Harappan brick ratio **4:2:1**; Harappan streets main 9–10 m, secondary 4–5 m, lanes 1.8–2.4 m.
- Mathematics at top of Vedāṅga Śāstras: "yathā śikhā mayūrāṇām... gaṇitaṁ mūrdhani sthitam" = like the crest of a peacock (SEE MCQ).

BOOK p.141 Historical evidence ↗

BOOK p.142 Salient features ↗

BOOK p.143 Zero ↗

BOOK p.144 Place value ↗

BOOK p.146 Decimal ↗

BOOK p.150 Measurements ↗

Great mathematicians (SEE 5a: "any four")

Name · date	Work	Contribution
Śulba Sūtra authors (Baudhāyana, Āpastamba, Kātyāyana) · 800–600 BCE	Śulba Sūtras	Earliest geometry; Pythagorean theorem; $\sqrt{2}$; $\pi \approx 3.088$; altar constructions.
Piṅgala · c. 300 BCE	Chandaḥśāstra	Binary system, combinatorics, Pascal's triangle.
Āryabhaṭa · 476–550 CE	Āryabhaṭīya	Place value, square/cube roots, sine table, $\pi = 3.1416$, kuṭṭaka, Earth rotates.
Varāhamihira · 505–587 CE	Pañcasiddhāntikā, Bṛhat Saṃhitā	Sine table, $\sin^2 + \cos^2$, combinatorics, magic squares.
Bhāskara I · 7th c.	Laghu/Mahā-bhāskarīya	Approx formula for sine; commentary on Āryabhaṭīya.
Brahmagupta · 598–668 CE	Brahmasphuṭasiddhānta	Zero and negative-number rules; algebra; quadratic; cyclic quadrilateral.
Mahāvīracārya · 800–870 CE	Gaṇita-sāra-saṅgraha	First dedicated maths textbook; permutations & combinations.
Bhāskara II · 1114–1185 CE	Līlāvati, Bijagaṇita, Siddhānta-śiromaṇi	Arithmetic & algebra; chakravāla; calculus ideas; $\pi = 3927/1250$.
Mādhava (Kerala) · c. 1350–1375	(Kerala school)	Infinite series for sin, cos, arctan, π (11 decimals), 250 years before Newton/Leibniz.

2.3 Unique ways to write numbers **IMPORTANT**

All three exploit poetry. **Rule: "Āṅkānām vāmato gatiḥ": digits are read right to left** (the first word is the units digit).

(a) Bhūta-saṃkhyā: numbers as objects / beings / concepts

Bhūta-saṃkhyā system	
Number	Represented by (partial list only)*
0	Śūnya, Pūrṇa, Kha
1	Ādi, Candra, Prithivī, Eka
2	Aśvin, Pairs Of Limbs, Ayana, Dvandva, Dvi
3	Rāma, Guṇa, Loka, Kāla, Agni, Trinetra
4	Veda, Śruti, Yuga, Āśrama, Varṇa, Samudra, Kṛta
5	Bhūta, Śāstra, Bāṇa, Pāṇḍava, Indriya
6	Aṅga, Rtu, Darśana, Śanmukha, Ṣaṭ
7	Rṣi, Adri, Svara, Dhātu, Chandas
8	Vasu, Bhujāṅga, Siddhi, Dik, Kuñjara, Nāga
9	Gṛha, Aṅka, Nanda
10	Dik, Aṅgulī, Avatāra, Rāvaṇaśira
11	Rudra
12	Āditya, Rāsi
13	Viśva, Kāma
14	Manu
15	Tithi, Dina

Bhūta-saṃkhyā values (your notes)

Phrase	Digits	Number
veda-vedāṅka-candrāḥ	4,4,9,1	1944
khādri-rāmāgnayaḥ	0,7,3,3	3370
netra-kha-bhūmi-śaśi-sarpa	2,0,1,1,8	81102
bhujaṅga-nanda-dvi-naga-aṅga-bāṇa-ṣaṭ-kṛta-indavaḥ	8,9,2,7,6,5,6,4,1	146,567,298

CIE-1 values: manu 14, agni 3, elephant 8, tithi 15, rudra 11, Indra 3, Pāṇḍava 5. Quick set: eye/ pair 2; Rāma, agni, guṇa 3; Veda, yuga 4; bhūta, bāṇa, Pāṇḍava, indriya 5; rtu, aṅga 6; ṛṣi, adri 7; vasu, nāga, gaja, bhujaṅga 8; graha, nanda 9; kha/śūnya 0.

Trap: the Unit-2 QB gives khādri-rāmāgnayaḥ = 1733. By the rule in your notes (0,7,3,3 read right to left) it is **3370**. Write 3370 with working.

(b) Kaṭapayādi: consonants carry digits (mnemonic)

- Letters from **ka** start: ka–ña = 1...9,0; ṭa–na = 1...9,0; pa–ma = 1...5; ya–la = 1...9. **Standalone vowels = 0; in a conjunct only the last consonant counts; trailing bare consonant is ignored.** Read right to left.
- Examples: **bhavati** = bha 4, va 4, ti 6 → **644**. **śaktyāloke** = 5-1-3-1 → **1315**. **āyurārogyasaukhyam** → 1,712,210.
- Famous: Śaṅkaravarman's *Sadratnamālā* verse "bhadraṃbudhisiddha..." encodes **$\pi = 3.14159265358979324$** (17 decimals). Also 72 *melakartā* rāgas: first two syllables give the rāga number (Mechakalyāṇī = 65).

- **Advantage:** long numbers become easy-to-remember words/verses, ideal for oral tradition and codifying results.

1	2	3	4	5	6	7	8	9	0
क	ख	ग	घ	ङ	च	छ	ज	झ	ञ
ट	ठ	ड	ढ	ण	त	थ	द	ध	न
प	फ	ब	भ	म	-	-	-	-	-
य	र	ल	व	श	ष	स	ह	ळ	-

Kaṭapayādi letter-value table

Kaṭapayādi System: Examples
72 Melakartas in Carnatic Music

मेचकल्याणि - मे = 5, च = 6; Therefore **65**.
वागदीक्षरी - वा = 4, ग = 3; Therefore **34**.
गानमूर्ति - गा = 3, न = 0; Therefore **3**.

1	2	3	4	5	6	7	8	9	0
क (ka)	ख (kha)	ग (ga)	घ (gha)	ङ (ṅa)	च (ca)	छ (cha)	ज (ja)	झ (jha)	ञ (ña)
ट (ṭa)	ठ (ṭha)	ड (ḍa)	ढ (ḍha)	ण (ṇa)	त (ta)	थ (tha)	द (da)	ध (dha)	न (na)
प (pa)	फ (pha)	ब (ba)	भ (bha)	म (ma)					
य (ya)	र (ra)	ल (la)	व (va)	श (śa)	ष (ṣa)	स (sa)	ह (ha)		

72 melakarta example

(c) Āryabhaṭa system

- Consonants ka...ma = 1...25 (varga); ya...ha = 30, 40, 50, 60, 70, 80, 90, 100 (avarga). The 9 vowels give place values: a = 10^0 , i = 10^2 , u = 10^4 , ṛ = 10^6 , ḷ = 10^8 , e = 10^{10} , ai = 10^{12} , o = 10^{14} , au = 10^{16} (steps of 100).
- Examples: **guṇa** = gu (3×10^4) + ṇa (15) = **30015**. **khyughr** = $(2+30) \times 10^4 + 4 \times 10^6 = \mathbf{4,320,000}$ (a Mahāyuga). (Your notes print "2×30", which is a typo; it is 2+30.)

क/k	ख/kh	ग/g	घ/gh	ङ/ṅ	च/c	छ/ch	ज/j	झ/jh	ञ/ñ
1	2	3	4	5	6	7	8	9	10
ट/ṭ	ठ/ṭh	ड/ḍ	ढ/ḍh	ण/ṇ	त/t	थ/th	द/d	ध/dh	न/n
11	12	13	14	15	16	17	18	19	20
प/p	फ/ph	ब/b	भ/bh	म/m					
21	22	23	24	25					
य/y	र/r	ल/l	व/v	श/ś	ष/ṣ	स/s	ह/h		
30	40	50	60	70	80	90	100		

अ / आ	इ / ई	उ / ऊ	ऋ / ॠ	ॡ	ए	ओ	ऐ	औ
a / ā	i / ī	u / ū	ṛ / ṛī	ḷ	e	o	ai	au
10^0	10^2	10^4	10^6	10^8	10^{10}	10^{12}	10^{14}	10^{16}

Āryabhaṭa letter values

Yuga cycle		
Yuga	Start (– End)	Length
<i>Krita (Satya)</i>	3,891,102 BCE	1,728,000 (4,800)
<i>Treta</i>	2,163,102 BCE	1,296,000 (3,600)
<i>Dvapara</i>	867,102 BCE	864,000 (2,400)
<i>Kali*</i>	3102 BCE – 428,899 CE	432,000 (1,200)
Years: 4,320,000 solar (12,000 divine)		

Yuga cycle (4,320,000 yrs)

Bhūta-saṃkhyā	Kaṭapayādi	Āryabhaṭa
Objects/concepts stand for digits	Consonants stand for digits; place value by position (right to left)	Each consonant = value; vowels = place value (powers of 100)
Uses whole words	Meaningful words from digits	Compact; used in Āryabhaṭiya

[BOOK p.146 Bhūta-saṃkhyā ↗](#) [BOOK p.148 Kaṭapayādi ↗](#)

2.4 Piṅgala & the binary system **IMPORTANT**

- **Chandaḥśāstra** by Piṅgala, c. **300 BCE** (textbook: 200–300 BCE; your Unit 2 notes: "2nd c. BCE"). Treatise on prosody (Chandas = Vedāṅga). Metre hierarchy: syllable → pāda → metre.
- Two syllable types: **Laghu (light/short, 1 mātrā)** and **Guru (heavy/long, 2 mātrās)**.
Guru has **4 varieties**: (1) long vowel; (2) short vowel before a conjunct consonant; (3) short vowel before anusvāra/visarga; (4) last syllable of a quarter (pāda).
- **Convention: Laghu = 1, Guru = 0** (textbook Table 6.9 & 8.7, Unit 2 notes, SEE Feb 2026 MCQ "1 and 0").

Conflict: Unit 3 notes and CIE-2 solution say Laghu = 0 / Guru = 1. For MCQs use **Laghu = 1, Guru = 0** (SEE + textbook). In a written answer, state your convention explicitly.

Gaṇas: 3-syllable binary words, mnemonic yamātārājabhānasalagām

TABLE 6.9 Eight Gaṇas Defined by Piṅgala

Sl. No.	Gana Name	Binary Word*
1	'Ya' Gana	100
2	'Ma' Gana	000
3	'Ta' Gana	001
4	'Ra' Gana	010
5	'Ja' Gana	101
6	'Bha' Gana	011
7	'Na' Gana	111
8	'Sa' Gana	110

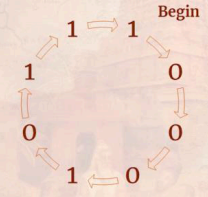
* 1 = Laghu; 0 = Guru

Textbook Table 6.9 (1 = Laghu, 0 = Guru)

Mnemonic for the 8 "gaṇas" of Piṅgala

S.No. Word	Gana Name	Binary
1	"Ya" Gana	100
2	"Ma" Gana	000
3	"Ta" Gana	001
4	"Ra" Gana	010
5	"Ja" Gana	101
6	"Bha" Gana	011
7	"Na" Gana	111
8	"Sa" Gana	110

यमाता-राज-भान-सलगम्
yamātā-rāja-bhāna-salagam



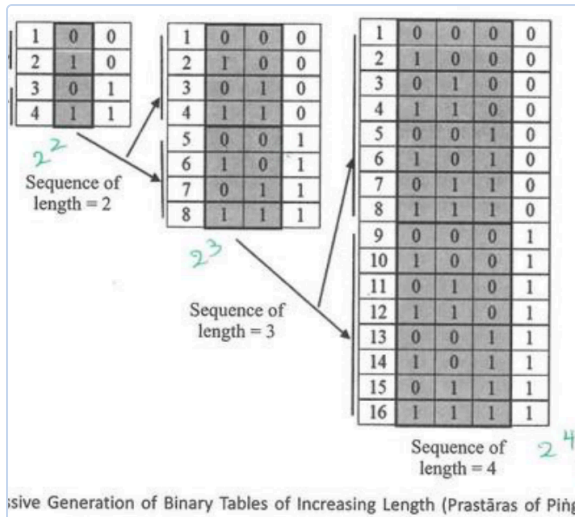
Mnemonic and binary cycle (your notes)

IMPORTANT Gaṇas and the de Bruijn sequence (SEE Feb 2026 4b)

- 8 gaṇas = all $2^3 = 8$ binary words of length 3: Ya 100, Ma 000, Ta 001, Ra 010, Ja 101, Bha 011, Na 111, Sa 110.
- The mnemonic *ya-mā-tā-rā-ja-bhā-na-sa-la-gam* converts to 1000101110; the last two digits "10" repeat the first two, so dropping them leaves a **cyclic binary cycle of length 8**: any 3 consecutive digits (cyclically) give a different gaṇa.
- This is exactly a **de Bruijn sequence B(2,3)**: every possible length-n string over a k-letter alphabet occurs exactly once as a substring ($k = 2$: laghu/guru; $n = 3$). Piṅgala did this ~2,300 years before de Bruijn (rediscovered recently). Applications: coding, robotics, cryptography.
- CIE-1: binary cycle of length 4 = 0000100110101111 (16 words).

Piṅgala's combinatorial toolkit

Term	Meaning
Prastāra	Listing all 2^n patterns as an array (built by doubling: copy the array, add a column with 0 in first half, 1 in second).
Samkhyā	Count of rows = 2^n .
Naṣṭa	Row number → pattern.
Uddiṣṭa	Pattern → row number.
Lagakriyā	How many patterns have r ones (nCr), via Varṇa-Meru = Pascal's triangle (rows 1; 1 1; 1 2 1; 1 3 3 1...), later rediscovered by Pascal (1655). Mātrā-meru gives Fibonacci-type counts.
Adhvayoga	Space the array occupies.



Prastāras of length 2, 3, 4 (textbook Fig 8.7)

BOOK p.154 Piṅgala & binary ↗

BOOK p.156 Gaṇas & de Bruijn ↗

BOOK p.210 Chandaḥśāstra maths ↗

BOOK p.212 Naṣṭa/Uddiṣṭa ↗

Worked algorithms (textbook p.212–213)

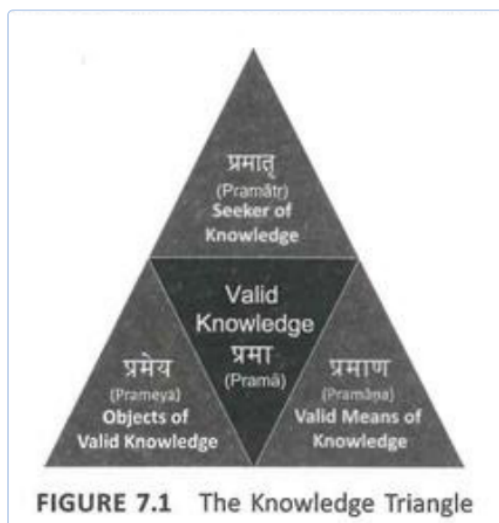
Naṣṭa: if the number is divisible by 2 write **1** and halve; if not, write **0**, add 1 and halve; repeat until length reached. Row 13 (length 4): 13→0, 7→0, 4→1, 2→1 ⇒ **0011**.

Uddiṣṭa: start at 1; scan from right; first '1' ×2; for each next digit: '1' → double, '0' → double and subtract 1. 0111 ⇒ 15.

Shortcut: row = (mirror of the binary word read as a number) + 1. 5th row of length 3 = 001 → mirror 100 = 4 → +1 = 5.

2.5 Knowledge framework: Prameya, Pramāṇa, Saṃśaya **IMPORTANT**

Indian epistemology (Pramāṇa Śāstra): knowledge = true awareness acquired by a mode of inquiry. Two schools handle it: **Vaiśeṣika** (ontology: *what exists*, i.e. Prameya) and **Nyāya** (logic & epistemology: *how to know validly*, i.e. Pramāṇa).



The Knowledge Triangle (textbook Fig 7.1)

IMPORTANT Knowledge Triangle (MQP 4a)

- **Pramātr:** the seeker/knower.
- **Prameya:** object of valid knowledge (what is to be known).
- **Pramāṇa:** valid means of knowledge.
- Centre: **Pramā** = valid knowledge produced.
- **Saṃśaya** (doubt/ambiguity) is where inquiry **begins** (MQP Q18). Then Prayojana → argument & logic → Siddhānta (tenet) → Nirṇaya (conclusion).

Prameya (Vaiśeṣika: physical reality) · founder Kaṇāda, ~370 sūtras

- Method: **Uddeśa** (enumerate) → **Lakṣaṇa** (define) → **Parīkṣā** (examine by perception & inference).
- **Six padārthas (categories):** **Dravya** (substance), **Guṇa** (attribute), **Karma** (action): objective/concrete; **Sāmānya** (universality/genus), **Viśeṣa** (specialty/species), **Samavāya** (inherence/combination): intellectual.
- **Nine dravyas:** Pṛthivī (earth), Ap (water), Tejas (fire), Vāyu (air), Ākāśa (ether), Dik (space), Kāla (time), Ātman (self), Manas (mind). 4 elements + mind finite and atomic; ākāśa, dik, kāla, ātman infinite.
- Atoms (*aṇu*): 2 aṇu = dvyaṇuka (dyad); 3 dyads = tryaṇuka (smallest visible).
- **Guṇa:** 17 listed by Kaṇāda, expanded to 24 (colour, taste, smell, touch, number, conjunction, cognition, pleasure...). **Karma** (5 types: throwing up, down, contraction, expansion, motion); action belongs to one substance, drives conjunction/disjunction.

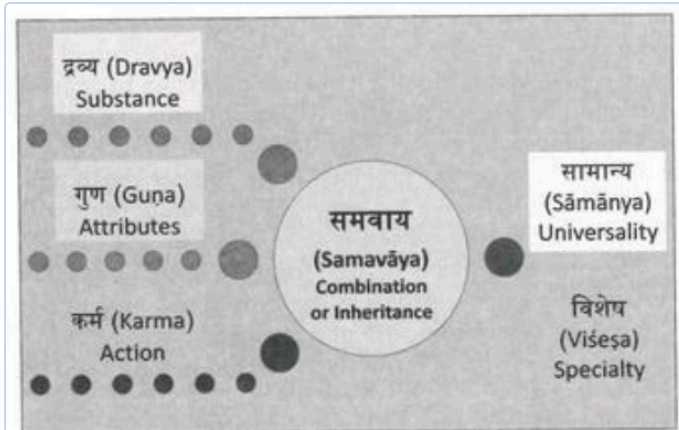


FIGURE 7.2 The Vaiśeṣika Framework

Vaiśeṣika framework (Fig 7.2)

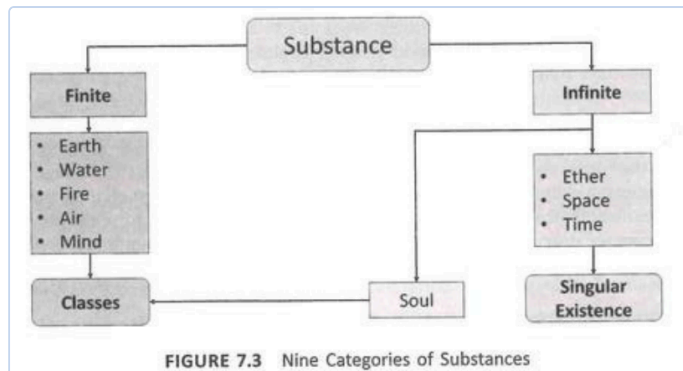


FIGURE 7.3 Nine Categories of Substances

Nine substances (Fig 7.3)

Pramāṇa: the four valid means (Nyāya) **IMPORTANT**

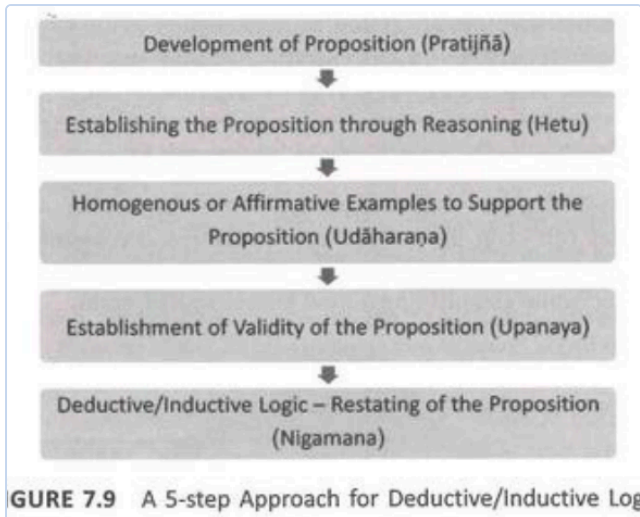
Pramāṇa	Meaning	Key points / example
Pratyakṣa (perception)	Direct knowledge via contact	5 senses + Manas (mind): sight, hearing, taste, smell, touch. Chain: Ātmā → Manas → Indriya → Artha. Most fundamental; all others rest on it. Types (CIE-1): laukika (ordinary) and alaukika (sāmānyalakṣaṇa, jñānalakṣaṇa, yogaja). Eg: counting 392 people.
Anumāna (inference)	Knowledge that "follows" perception	Smoke on hill → fire. Terms: Pakṣa (locus), Sādhya (what is to be proved), Hetu (reason), Vyāpti (invariable concomitance). 3 kinds: pūrvavat (a priori), śeṣavat (a posteriori), sāmānyatodrṣṭa (commonly seen).
Upamāna (analogy)	Comparison / similarity	Gavaya looks like a cow → recognise a nilgai. Types: sādrśya (similarity), vaidharmya (dissimilarity), asādhāraṇa-dharma (special trait: rhinoceros horn).
Śabda (testimony)	Word of a reliable person (āpta)	<i>āptopadeśaḥ śabdaḥ</i> (Nyāya Sūtra 1.1.7). Āpta = right knowledge, willing to share, no intention to deceive, free from attachment/aversion. Eg: doctor's advice; Vedic injunctions.

IMPORTANT Pañca-avayava anumāna (5-step inference) (SEE Feb 2026 4a)

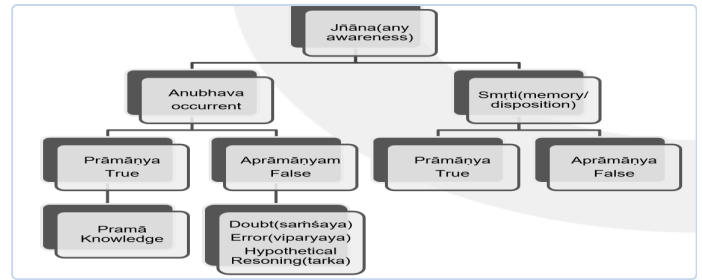
Example from your notes: "Force has been imparted to this ball because it is moving."

#	Step	Statement
1	Pratijñā (proposition)	The ball has the property of being subjected to force.
2	Hetu (reason)	Because it has motion.
3	Udāharaṇa (example)	Wherever there is motion there is force, like a cart.
4	Upanaya (application)	This ball is also so (has motion).
5	Nigamana (conclusion)	Hence the ball is subject to force.

Classic: Pratijñā "the hill has fire"; Hetu "because it has smoke"; Udāharaṇa "like a kitchen"; Upanaya "this hill has smoke"; Nigamana "so it has fire."



5-step logic (Fig 7.9)



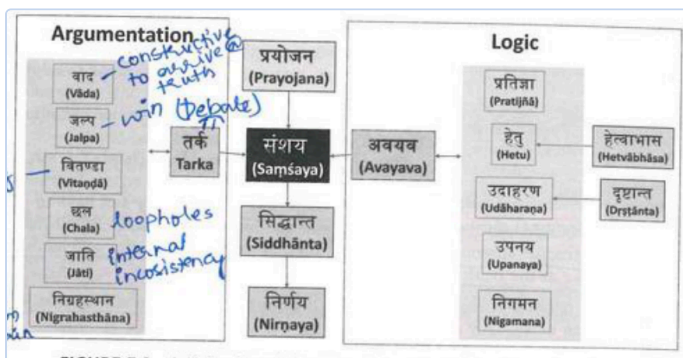
Jñāna tree: Anubhava/Smṛti → Pramā / Apramā (doubt, error)

Samśaya: ambiguity as the start of knowledge

- New knowledge begins by recognising ambiguity in existing knowledge. Five causes (textbook Fig 7.7): (1) several observed common properties; (2) uncommon/unknown properties with no explanation; (3) **conflicting findings**; (4) variation in results; (5) **non-apprehension** of the problem.

Framework for establishing valid knowledge (Nyāya)

- Tarka** = debate/reasoning in presence of ambiguity. Between a **vādin** (proponent) and **prativādin** (opponent). Six argument modes: **Vāda** (constructive, truth-seeking) · **Jalpa** (win the debate) · **Vitaṇḍā** (only attack, "I know you are wrong") · **Chala** (exploit loopholes) · **Jāti** (point to inconsistency) · **Nigrahasthāna** (raise objections / defeat points).
- Hetvābhāsa**: five fallacies: **Savyābhicāra** (erratic), **Viruddha** (contradictory), **Prakaraṇasama** (restates the problem; circular), **Sādhya-sama** (unproved), **Kālātīta** (mistimed / irrelevant).
- Siddhānta** = established tenet (4 kinds in Nyāya); **Nirṇaya** = final conclusion.



Nyāya framework (Fig 7.8; blue ink is from your book)



Five fallacies (Fig 7.10)

- BOOK p.162 Tarka ↗
- BOOK p.163 Knowledge triangle ↗
- BOOK p.165 Prameya ↗
- BOOK p.167 Dravyas ↗
- BOOK p.170 Guṇa & Karma ↗
- BOOK p.173 Pramāṇa ↗
- BOOK p.176 Samśaya ↗
- BOOK p.178 Framework ↗
- BOOK p.181 Deductive logic ↗
- BOOK p.182 Fallacies ↗

UNIT 3 · Science, Engineering & Technology in IKS

3.1 Mathematics overview

- Gaṇita** = science of computation (arithmetic, algebra, geometry, astronomy). Features: **algorithmic approach**, **sūtra-style verses**, **integration with other sciences**.
- Phases: **Vedic** (1500–500 BCE: Śulba Sūtras) → **Classical** (500 BCE–1200 CE: Āryabhaṭa, Brahmagupta, Bhāskara I & II) → **Kerala School** (1300–1600: Mādhava, Nīlakaṇṭha).
- Arithmetic**: Brahmagupta's rules: $a+0 = a$; $a-0 = a$; $a \times 0 = 0$; $a/0$ undefined; $(+)\times(+) = +$, $(+)\times(-) = -$, $(-)\times(-) = +$. Dhana = fortune (+), ṛṇa = debt (-). Multiplication *bhāvanā* (456×23 : $456 \times 3 = 1368$; $456 \times 20 = 9120$; sum 10488). Āryabhaṭa's square-root method (digit pairs); **Bakhshālī** square-root approximation (~200 CE). Series: $\Sigma n = n(n+1)/2$; $\Sigma n^2 = n(n+1)(2n+1)/6$; $\Sigma n^3 = [n(n+1)/2]^2$. AP: $a_n = a + (n-1)d$ (CIE-2 Q).

Geometry (Rekhā-gaṇita), Śulba Sūtras **IMPORTANT**

- Constructive and application-oriented (altars, temples, towns), unlike Greek axiomatic. Rope (śulba) constructions: square, rectangle, circle; shape transformations (square ↔ rectangle ↔ circle of equal area; doubling a square).
- Baudhāyana theorem (c. 800 BCE)**: "the diagonal of a rectangle produces both areas which its length and breadth produce separately" = Pythagorean theorem, centuries before Pythagoras. Triplets (3,4,5), (5,12,13).
- $\sqrt{2} \approx 1 + 1/3 + 1/(3 \times 4) - 1/(3 \times 4 \times 34) = 1.4142156$.
- Shadow geometry with gnomon (śaṅku): $h_1/s_1 = h_2/s_2$ (similar triangles).

TABLE 8.3 History of Approximations to π by Indian Mathematicians

	Value of π	Accuracy (Decimal places)	Method
Śulba-sūtras (around 800 BCE)	3.08888	1	Geometrical
Jaina texts (500 BCE)	$\sqrt{10} \approx 3.1623$	1	Geometrical
Āryabhaṭa (499 CE)	$\frac{62832}{20000} \approx 3.1416$	4	Polygon doubling ($4.2^8 = 1024$ sides)
Bhāskarācārya (Līlāvati)	$3927/1250 \approx 3.1416$	4	Polygon doubling
Mādhava (1375 CE)	$\frac{2827433388233}{9 \times 10^{11}} \approx 3.141592653592\dots$	11	Infinite series with end corrections
Ramanujan (1914 CE)		17 million	Modular equation

History of π (textbook Table 8.3)

IMPORTANT Value of π

- Early Vedic ≈ 3 ; Śulba ≈ 3.088 ; Jain texts $\sqrt{10} \approx 3.1623$; **Āryabhaṭa 62832/20000 = 3.1416** (4 decimals); Bhāskara II 3927/1250; **Mādhava ≈ 3.14159265359 (11 decimals)**; Ramanujan 17 million places.
- Mādhava series: $\pi/4 = 1 - 1/3 + 1/5 - 1/7 + \dots$

MQP Q32 ("approx value of π used in ancient India": $3.14 / 3 / 22/7 / \sqrt{10}$): textbook says early works used 3 and Jain works $\sqrt{10}$. If asked "ancient" with no other clue, textbook p.206 supports both **3** and **$\sqrt{10}$** ; pick whichever fits the options. 22/7 is not an Indian-text value here.

Trigonometry (Jyotpatti), Algebra (Bijagaṇita)

- India shifted from Greek **chord** to **half-chord = jyā (sine)**; *kojyā* = cosine. Āryabhaṭa: table of sine differences, 24 intervals of 3.75° ($0-90^\circ$), $R = 3438'$. Recursive sine computation; Nīlakaṇṭha finite differences; Mādhava's series for sin, cos, arctan.
- Word history: **jyā** → **Arabic jiba/jaib** → **Latin sinus** → "**sine**".
- Algebra**: "science of seeds" (*bīja*). Āryabhaṭa: **kuṭṭaka** (pulveriser) for $ax + c = by$. Brahmagupta: quadratic, zero and negatives (*dhana/ṛṇa*), *Brāhmasphuṭa Siddhānta* (628 CE). Bhāskara II: **chakravāla** (cyclic method) for Pell-type $Nx^2 + 1 = y^2$ ($61x^2 + 1 = y^2$: $x = 226153980$). Symbols: *yu* (add), *kṣa* (subtract), *gu* (multiply), *bhā* (divide), *mū* (root).

[BOOK p.194 Great mathematicians ↗](#)

[BOOK p.204 Geometry ↗](#)

[BOOK p.206 \$\pi\$ ↗](#)

[BOOK p.207 Trigonometry ↗](#)

[BOOK p.209 Algebra ↗](#)

[BOOK p.214 Magic squares ↗](#)

3.2 Astronomy & the Indian calendar IMPORTANT

Time units

Year	Basis	Length
Sāvana	Sunrise to sunrise	366 sāvana days
Saura (solar)	Sun's one zodiac round; 12 rāsis (≈30.4 days each)	≈ 365.2564 days
Cāndra (lunar)	Moon; 12 māsas of ≈29.5 days	≈ 354.367 days

- Solar–lunar gap ≈ 11 days/yr → **adhika māsa (intercalary month) about every 32–33 months (≈2.5 yrs)**. A 5-year small **yuga** aligns the two. **Mahāyuga** = 4 yugas (Kṛta, Tretā, Dvāpara, Kali) = **4,320,000 years**.
- Lunar month: Śukla pakṣa (waxing) + Kṛṣṇa pakṣa (waning); between two Amāvāsyā (new moon) or Pūrṇimā. Months named after the nakṣatra of the full moon (Caitra, Vaiśākha...).
- Tithi**: Moon–Sun angular distance grows by **12°**; 30 tithis (15 + 15); Amāvāsyā 0°, Pūrṇimā 180°; duration varies roughly 19–26 h (your notes print "26 h 47 min to 10 h 59 min").

IMPORTANT **Pañchāṅga: five limbs** (SEE Feb 2026 6a; MQP Q25: *Rāśi is not a limb*)

Element	Meaning	Detail
Tithi	Lunar day	12° Sun–Moon separation; fixes festivals/fasts.
Vāra	Weekday	7 days: Ravi, Soma, Maṅgala, Budha, Guru, Śukra, Śani (Sun, Moon, Mars, Mercury, Jupiter, Venus, Saturn).
Nakṣatra	Lunar mansion	27 divisions, 13°20' each (360/27); Moon stays ≈1 day in each.
Yoga	Sun + Moon longitudes	Sum ÷ 13°20' → 27 yogas ; auspicious or not.
Karaṇa	Half a tithi (6°)	11 karaṇas: 4 fixed + 7 repeating.

Uses: muhūrta (auspicious times), festivals, agriculture, navigation. Solar (Sauramāna: Tamil Nadu, Kerala) vs lunar (Cāndramāna: most of India).

As science + culture (MQP 6a): computed from Sūrya Siddhānta/Āryabhaṭṭīya with predictions of eclipses, equinoxes, solstices (scientific document); guides rituals, farming, social life (cultural guide).

Āryabhaṭṭīya (499 CE) & Siddhāntic tradition IMPORTANT

- Siddhānta** = established doctrine/scientific treatise. Āryabhaṭṭīya: **121 verses, 4 sections**: **Gītikāpāda** (time cycles, revolutions), **Gaṇitapāda** (maths: roots, areas, sine table), **Kālakriyāpāda** (time, calendar), **Golapāda** (sphere, eclipses, diurnal motion).
- Ideas: **Earth is spherical and rotates on its axis** (day and night; boat analogy: stationary stars appear to move backwards); eclipses are due to shadows (not demons); Moon shines by reflected light; sine function; planetary mean and true positions with **manda** and **śighra** corrections. Influence on later Indian, Arab and European astronomers.
- Siddhāntic scholars: Varāhamihira (Pañcasiddhāntikā: Pitāmaha, Vasiṣṭha, Romaka, Pauliśa, Sūrya), Bhāskara I, Brahmagupta, Lalla, Bhāskara II. Mean of 5 siddhāntas combined Indian, Greek, Babylonian ideas.

Yantras & Jantar Mantar

Yantra	Use
Śaṅku (gnomon)	Shadow length; noon, direction, latitude
Gola (armillary sphere)	Celestial coordinates; motion of stars
Nāḍivalaya	Time since sunrise, lagna (60 ghaṭikā, 12 signs)
Cakra	Sun's altitude; planetary longitude/latitude
Turiya	Quadrant; altitude and zenith distance
Ghaṭī (water clock)	Time intervals (CIE-2: the yantra for time)
Phalaka, Dhī	Altitude/zenith plank; plumb-line inclination

Jantar Mantar: Sawai Jai Singh II (1686–1743): Delhi, Jaipur, Ujjain, Varanasi, Mathura. Instruments: Samrāt Yantra, Rāma Yantra, Jaya Prakāśa. Jyotiṣa = astronomy + astrology, a Vedāṅga; source text Vedāṅga Jyotiṣa (SEE MCQ 1.29).

[BOOK p.223 Unique aspects ↗](#)

[BOOK p.230 Calendar elements ↗](#)

[BOOK p.234 Āryabhaṭīya ↗](#)

[BOOK p.237 Pañchāṅga ↗](#)

[BOOK p.241 Yantras ↗](#)

[BOOK p.244 Jantar Mantar ↗](#)

3.3 Metals & metalworking **IMPORTANT**

- **Mining/extraction:** sites Zawar, Rajpura-Dariba, Khetri, Chitradurga, Singhbhum. Steps: prospecting → mining (open, adit, shaft; Zawar ≈500 ft) → fire-setting + chiselling → ore dressing → roasting → smelting (charcoal + flux) → refining.
- **Zinc:** India produced metallic zinc first on industrial scale, by **downward distillation at Zawar (9th–13th c. CE)**, ≈1000 °C; exported; used for brass.
- **Copper & alloys:** Copper (Khetri, Singhbhum, Chitradurga); **bronze = Cu + Sn**, **brass = Cu + Zn** (calamine); Copper was the first metal used (CIE-2).
- **Gold** by panning + mercury amalgamation; **silver** from galena by roasting and cupellation; **mercury** from cinnabar (Rasa-sindūra) by distillation; Ayurvedic bhasmas.
- **Apparatus (yantras):** Mūṣā (crucible), Pātana (distillation), Dhelkī (mercury), **Vāluka (sand bath: uniform heating)**, Dolā (suspension), Dhūpa (fumigation). Furnaces up to 1400 °C.

IMPORTANT Wootz steel (CIE-2, MQP 6b)

- High-carbon (**1–1.5% C**) crucible steel, South India (Tamil Nadu, Karnataka, Telangana), from ≈700 BCE (textbook: wootz from 4th c. BCE; CIE-2: 300 BCE–1700 CE). Name from Tamil/Kannada *ukku*.
- **Process:** wrought-iron pieces + carbon material (wood/leaves) sealed in clay crucibles, ≈1400 °C for ~6 h; carbon diffuses in; slow cooling forms carbides → **watered/damask pattern**. (European cementation took 6–20 days.)
- **Properties:** hard yet tough, razor-sharp, flexible, patterned.
- **Global value:** exported to Middle East/Persia/Europe; basis of legendary **Damascus swords**. Faraday studied it; precursor of alloy steels.
- **Decline (19th c.):** colonial policies/taxes, cheap European industrial steel, loss of traditional knowledge (≈150 years).

IMPORTANT Delhi Iron Pillar (SEE Feb 2026 6b)

- 5th c. CE; **7.2 m high, >6 tonnes, 99.7% pure wrought iron**; corrosion-free for 1,600 years.
- Why: low phosphorus + high purity; forms passive protective layer of **iron hydrogen phosphate**. Shows forge-welding, temperature and composition control.
- Other: Dhar pillar (12th c., twice as big), Konark iron beams (29), Puri beams, Thanjavur cannon, Suśruta's >100 steel surgical tools.

Iron type (Rasaratna-samuccaya)	Property	Use
Kānta-loha	Soft, magnetic, ductile	Tools, utensils
Tikṣṇa-loha (steel)	Hard, sharp, temperable	Swords, weapons
Muṣṭa-loha (cast iron)	Brittle, low melting	Casting

- **Steel making:** carburisation of wrought iron (crucible) and decarburisation of cast iron (→ mild steel). Iron extraction known from 1200 BCE; ayurvedic extraction from biotite using kājī, triphala, cow urine/milk, heated to ≈1400 °C.

Lost-wax casting (*madhucchiṣṭa-vidhāna*, "cire perdue") **IMPORTANT**

1. Wax model (beeswax + oils/resin) with gates and vents → 2. Coat with clay (+ rice husk, sand, dung) layers, dry → 3. **Dewax:** heat, wax melts out, mould hardens → 4. Pour molten metal (bronze, brass, **pañcaloha** = Au, Ag, Cu, Sn, Zn) → 5. Break the mould; finish (chisel, polish, patina).

Each casting is unique (mould destroyed). Examples: Harappan "**Dancing Girl**" (Mohenjo-Daro, 2500 BCE), **Chola bronzes** (9th–12th c.; Naṭarāja), Bastar, Odisha. Somesvara's *Mānasollāsa* (12th c.) describes it. Still done at Swamimalai. Modern use: jewellery, turbine blades.

[BOOK p.253 Wootz ↗](#)

[BOOK p.255 Mining ↗](#)

[BOOK p.259 Zinc ↗](#)

[BOOK p.265 Iron & steel ↗](#)

[BOOK p.271 Steel manufacture ↗](#)

[BOOK p.272 Lost wax ↗](#)

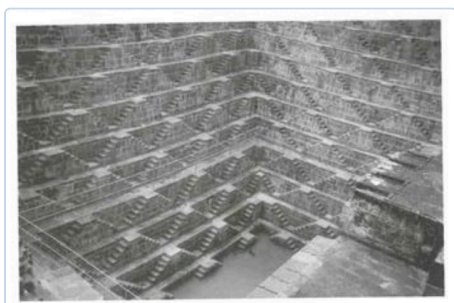
[BOOK p.273 Apparatus ↗](#)

3.4 Dyes & painting

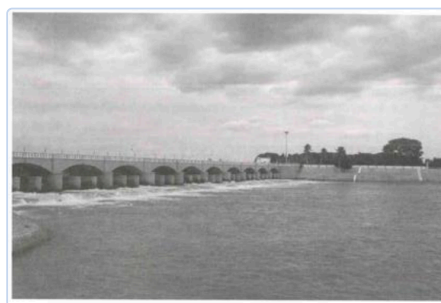
- Sources: **plants** (indigo = blue, madder = red, turmeric = yellow, saffron), **minerals** (ochres, malachite, azurite), **insects** (lac, cochineal), metals (orpiment). **Mordants** (alum, iron salts, lime, tannins) fix colour; Mohenjo-Daro dyed cotton fragment proves mordant knowledge.
- Methods: direct dyeing; mordant dyeing (madder + alum = red, + iron = purple); **resist dyeing** (Bandhani, Ikat); overdyeing (turmeric + indigo = green). **Indigo** (*Indigofera tinctoria*): leaves fermented to indigotin; India main supplier till 19th c. (MQP Q33: leaf).
- Painting: murals (Ajanta, Ellora, Sittanavasal, Bagh) on lime plaster; miniatures; fresco; folk (Madhubani, Warli, Pattachitra). Binders: gum arabic, resins, egg white, casein. Words *calico*, *chintz*.

3.5 The art of making perfumes (*Gandhaśāstra* / *Gandhayukti*) **IMPORTANT**

- **Sources:** flowers (jasmine, rose, champaka), woods (sandal, agarwood), herbs (vetiver, camphor), resins (benzoin), spices, animal (musk, ambergris).
- **Methods:** **deg-bhapka steam distillation** (copper still *deg* + receiver *bhapka*; still used for **attar at Kannauj**); maceration/infusion in oil or fat; distillation of woods/resins; blending with **fixatives** (musk, sandal oil); ageing.
- **Products:** sugandhi taila, attar, dhūpa/agarbatti, gandha-chūrṇa, rose/vetiver/saffron water.
- **Texts:** Varāhamihira's *Brhat Samhitā* chapter 77 "**Gandha-yukti**" (37 verses; uses permutations: 16 ingredients choose 4 = 1,820 combos × 96 = 174,720 perfumes); Gandhasāra; Charaka/Suśruta. 8-step process *karmāṣṭakam*: śodhana, vāsana, virecana, bhāvanā, pāka, bodhana, dhūpana, vedhana.
- **Significance:** ritual, medicine, cosmetics, status/hospitality; global trade; "attar" ← Sanskrit *gandha* via Persian *itr*.



Chand Baori stepwell (textbook Fig 11.3)



Grand Anicut / Kallanai, world's oldest dam still in use

[BOOK p.294 Dyes & painting ↗](#)

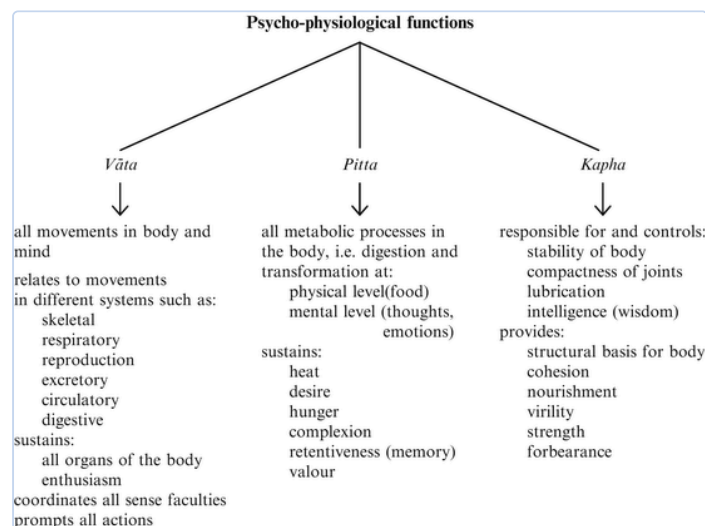
[BOOK p.295 Perfumes ↗](#)

UNIT 4 · Yoga, Ayurveda, Water, Town Planning & Architecture

4.1 Tridoṣa: Vāta · Pitta · Kapha **IMPORTANT**

	Vāta (movement)	Pitta (transformation)	Kapha (cohesion)
Elements	Air + Ether	Fire + a little Water	Water + Earth
Governs	Motion, nerve impulses, circulation, respiration, elimination	Digestion, metabolism, body heat, vision, enzymes/hormones , intellect "fire"	Structure, lubrication, immunity, growth, memory stability
Qualities	Light, dry, cold, mobile, rough	Hot, sharp, light, liquid, slightly oily	Heavy, slow, cool, oily, smooth, stable
Sites	Colon, pelvis, ears, bones, skin	Small intestine, liver, eyes, blood	Stomach, chest, head, joints, fat
Dominant	Old age; autumn	Adulthood; summer	Childhood; spring
Aggravation	Anxiety, insomnia, dryness, gas	Irritability, acidity, heat, inflammation	Lethargy, heaviness, congestion, weight gain
Balance with	Warm moist meals, routine, oil massage (abhyāṅga)	Cooling foods, śītalī, calm	Light, warm, pungent food, vigorous exercise

- **Prakṛti** = congenital dosha constitution; **Vikṛti** = current imbalance (diet, lifestyle, season, stress). Pillars: Agni (digestive fire), Dhātu, Mala, Ojas, Srotas.
- Subtypes: Prāṇa, Udāna, Samāna, Vyāna, Apāna (Vāta); Pācaka, Rājaka (Pitta); Kledaka, Avalambaka (Kapha).
- **Physical + psychological health (MQP 7a)**: Pitta excess ↔ rajasic anger/perfectionism; Vāta ↔ rajas/tamas anxiety/fear; Kapha ↔ tamas inertia. Balanced doṣas → clear mind.
- Routines: Dinacharya (daily: sunrise, tongue cleaning, abhyāṅga, yoga, main meal at midday) and R̥tucharya (seasonal).



Psycho-physiological functions of the doṣas

IMPORTANT **Pitta doṣa and physiology** (SEE Feb 2026 7a): Fire-dominant principle of transformation; drives digestion and metabolism (agni), thermoregulation, vision, skin colour/pigmentation, enzymatic and hormonal action, intellectual sharpness. Sites: small intestine, liver, eyes, blood. Imbalance: acidity, irritability, inflammation; remedy: cooling diet and lifestyle.

[BOOK p.341 Ayurveda ↗](#)

[BOOK p.345 Tri-doṣas ↗](#)

[BOOK p.348 Dinacharyā ↗](#)

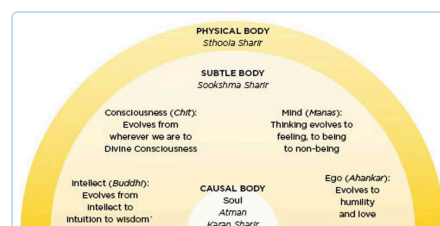
4.2 Triguna system **IMPORTANT**

Guṇa	Nature	Mental traits	Increase by
Sattva	Clarity, harmony, light	Calm, compassion, discernment, joy	Meditation, seva, fresh clean diet, ethics, satsang
Rajas	Activity, passion	Desire, ambition, restlessness, anger	Spicy/caffeine, overwork, competition
Tamas	Inertia, obscurity	Lethargy, confusion, apathy	Stale/heavy food, oversleeping, numbing media

- All three always present; dominance shapes behaviour (Sāṃkhya/Yoga; Gītā ch. 14 & 17). Triguna (Sattva, Rajas, Tamas) ≠ Tridoṣa. MQP Q45 answer: A.
- Link:** Doṣas = somatic/functional; Guṇas = psychological tone. Vāta ↔ rajas (tamas: fear-freeze), Pitta ↔ rajas, Kapha ↔ tamas (sattva when balanced).

4.3 Body–Mind–Intellect–Consciousness (BMI-C) **IMPORTANT**

- Body (Śarīra):** physical instrument; senses and action organs. **Mind (Manas):** emotions, sensory processing, likes/dislikes. **Intellect (Buddhi):** discrimination, reasoning, decisions. **Consciousness (Ātman/Puruṣa):** witnessing awareness.
- Flow:** Senses → Mind → Intellect → Action (Body), with Consciousness as the witness. Feedback: body state affects mind clarity; intellect can retrain mind and body (**Body is first enhanced by sleep and nutrition**).
- Pañcakośa:** Annamaya (body), Prāṇamaya (vital energy), Manomaya (mind), Vijñānamaya (intellect), Ānandamaya (bliss).
- Therapeutic chain: clear srotas + steady Agni → nourished dhātus → resilient body → calm mind → sharp intellect → sattvic insight.



Layers of the individual
(Body/subtle/causal)

[BOOK p.359 Tri-guṇa ↗](#)

[BOOK p.362 BMI-C complex ↗](#)

[BOOK p.363 Consciousness ↗](#)

[BOOK p.367 Five layers ↗](#)

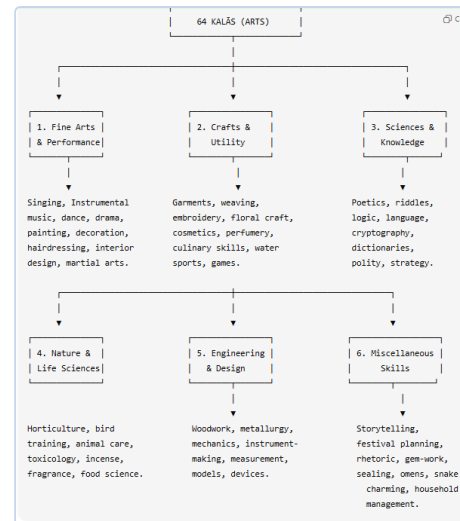
4.4 The 64 Kalās **IMPORTANT**

- Kalā* = art/skill that makes a person happy and skilled; **catuḥṣaṣṭi-kalā** = holistic curriculum of life skills (arts, crafts, sciences, social skills). Krishna learned all 64 under Sāṇḍīpani. Most famous list: **Vātsyāyana's Kāma Sūtra** (also Devī Bhāgavata Purāṇa, Kādambarī; Yaśodhara's *Jayamaṅgalā* commentary).
- Textbook grouping (Table 11.4, total 64): artistic 8 · cleaning/beautifying body 9 · decoration 6 · culinary 3 · play/entertainment 8 · intellectual pursuits 10 · engineering & technical 14 · special uncommon 6. Notes' grouping: Fine arts & performance · Crafts & utility · Literary/intellectual · Nature & life sciences · Engineering & design · Miscellaneous. Vāstu-vidyā is under **engineering/craftsmanship**; regional languages under **literary/intellectual**.
- Modern education relevance (SEE 8a):** multidisciplinary, skills + values, vocational competence, creativity, theory with practice, self-reliance: matches NEP 2020's holistic education, skill-based learning, and "learning by doing".

TABLE 11.4 A Summary of 64 Kalās

Type of Skills	No.
Artistic Skills: Singing, dancing, painting, theatrics, playing instruments, etc.	8
Skills in Cleaning and Beautifying Body: Dressing, decorating, painting the face, applying aromatics, combing hair, setting ornaments, etc.	9
Skills of Decoration: Making a covering of flowers for a bed, making the groundwork of jewels, covering the bed, applying an admixture of colors, etc.	6
Culinary Skills: Preparing delicious food items and drinks.	3
Play and Entertainment Skills: Playing on music in water, splashing with water, juggling, playing with thread, riddles, gambling, using children's toys, etc.	8
Skills for Intellectual Pursuits: Conversation, practicing language, composing, designing a literary work, lexicography and metres, reciting books, etc.	10
Engineering and Technical Skills: Needle works and weaving, spinning, carpentry, engineering, testing silver and jewels, metallurgy, mineralogy, practicing medicine, mechanics, etc.	14
Special Uncommon Skills: Knowing the mode of fighting of lambs, and birds, knowing conversation between male and female cockatoos, talking with fingers, prediction by heavenly voice, etc.	6

Textbook Table 11.4



64 Kalās classification (notes)

BOOK p.302 64 Kalās ↗

BOOK p.303 Table 11.4 ↗

4.5 Irrigation & water management **IMPORTANT** (not in your unit notes; from textbook)

- **Harappan/Dholavira:** 16 large reservoirs (~10% of city area); stepwell 73.4 × 29.3 × 10 m (3× Great Bath).
- **Mauryan:** Sudarśana dam (Girnar; Candragupta Maurya, improved under Aśoka/Rudradāman); Sanchi-area dams (300–200 BCE) with spillways; **Ahar–Pyne** system (Bihar; **pynes** = channels from hill rivers, **ahars** = embanked catchments) works as irrigation + flood control.
- **South India:** **tanks (ēri)** are the signature system (MQP Q46). **Grand Anicut/Kallanai** (Karikāla Chola, 1st c. CE, Kāveri) oldest dam still in use. Pallavas = great tank builders; Chola village assemblies ran committees: **Ēri-vāriyam** (supervision of tanks), **Kaliṅgu-vāriyam** (sluice gates).
- **Stepwells (baolis/vāvs)** of Gujarat/Rajasthan (Chand Baori): store water against seasonal variation; Aparājita-prcchā gives design; **Rajasthan johads/khadins** revived aquifers.
- **Sustainability:** rainwater harvesting, check dams/percolation tanks, contour bunds, community maintenance funded by local taxes. Modern parallel: **drip** (root-zone, water-efficient) and **sprinkler** irrigation (CIE-2: uniform distribution; less wastage than flood).

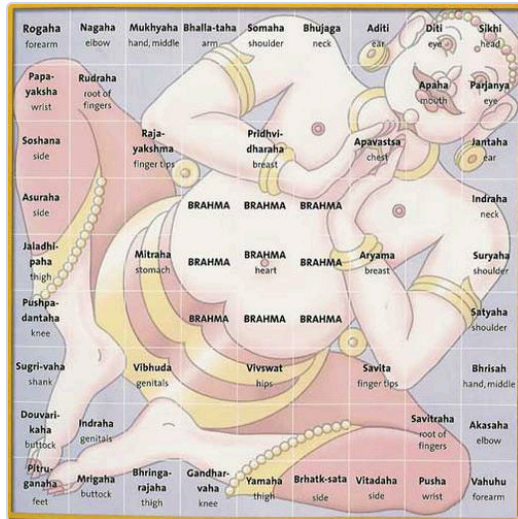
BOOK p.283 Irrigation in South India ↗

BOOK p.292 Irrigation & water mgmt ↗

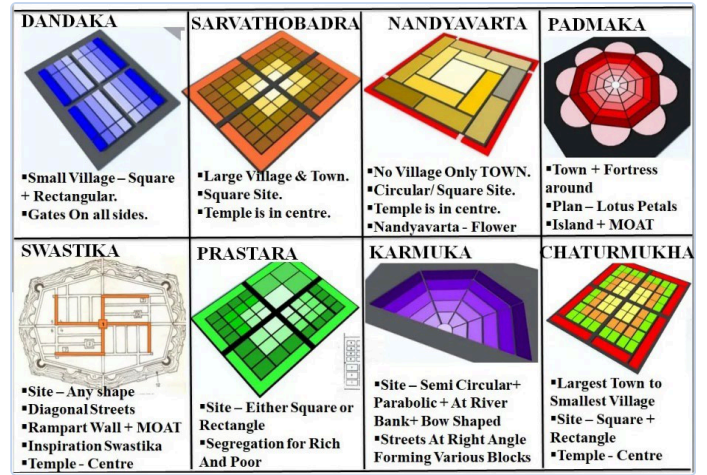
BOOK p.293 Anaicut & stepwells ↗

4.6 Vāstu Śāstra & architecture **IMPORTANT**

- **Vāstu Śāstra** = science of architecture/spatial design aligning buildings with **Pañca-mahābhūta**, 10 **Dikpālas**, **Vāstu-Puruṣa-Manḍala** and prāṇa flow. Texts: **Mānasāra**, Mayamata (MCQ: Mānasāra), Samarāṅgaṇa-sūtradhāra. Basic unit of town plan: **Pada**.
- **Principles:** orientation; Vāstu-Puruṣa-Manḍala (8×8 = 64 or 9×9 = 81, 10×10 grids); energy balancing; proportion (māna, āyatana, tāla-māna); site selection (soil, slope); environment (sun, wind); functional zoning. **East** most auspicious in Vāstu (MQP Q43).
- Manḍala types: Daṇḍaka, Sarvatobhadra, Nandyāvarta, Padmaka, Svastika, Prastara, Karmuka, Caturmukha (town layouts).



Vāstu-Puruṣa-Maṇḍala



Town-plan maṇḍala types

IMPORTANT Eight limbs of Vāstu (Aṣṭāṅga) (CIE-2; MQP Q48 answer "Aṣṭāṅga")

Your notes (exam list)	Meaning	Textbook Fig 12.4 list
1 Sthāpatya	Architecture: planning towns/temples	Yajamāna (host/owner)
2 Vāstu	Site selection, soil, topography	Sthāpatya (architecture)
3 Prāsāda	Building science: dimensions, materials	Śilpin (technicians: Sthapati, Sūtragrāhin, Vardhaki, Takṣaka)
4 Yāna	Vehicles/chariots	Bhūmi (land)
5 Śayana	Furniture and beds	Vāstospati (offerings/rituals)
6 Āsana	Seats, thrones	Pādavyāsa (site layout)
7 Āyudha	Weapons / protective structures	Vāstu (materials)
8 Ābharaṇa	Ornament, carving, colour	Alaṅkāraṇa (decoration)

Two lists exist. Your notes' list (Sthāpatya → Ābharaṇa) is what the QB/SEE-style MCQs use (QB: "not a limb = Yantra"). The textbook's Figure 12.4 gives the second column. Write the notes' list first; mention the textbook list as the "eight aspects of the construction process". CIE-2's key mentions Dik-nirṇaya and Bhūmi-parīkṣā (orientation, land examination) as examples; both are covered by Vāstu/Bhūmi.

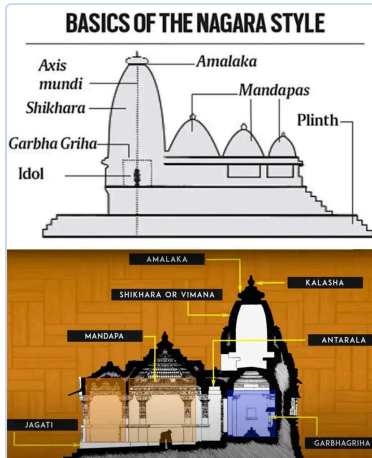
Town planning

- **Harappan:** earliest scientific planning: **grid, right-angle streets, citadel + lower town zoning, covered drains and soak pits**, granaries, public baths, Lothal dockyard, standard bricks; hygiene and utilities (MQP Q40, 44: sanitation and drainage; advanced water management).
- **Vedic/post-Vedic:** maṇḍala layouts; settlements: durga, pura/pattana, nagara, grāma. **Mauryan–Gupta:** Pāṭaliputra palaces, gardens, reservoirs, walls. **Medieval:** temple towns (Kāñchī, Madurai); Jaipur on Śilpa-śāstra (9×9 grid).
- **Arthaśāstra on towns:** emphasis on administration, security (forts, moats) and zoning (MQP Q41: C).
- **Principles:** maṇḍala layout; central temple/palace; zoning; water management (tanks, stepwells); defence; ecological orientation.

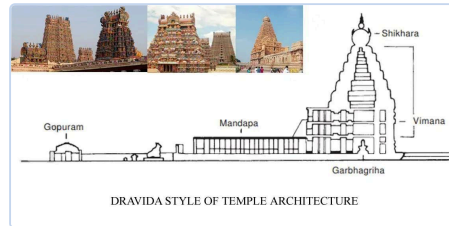
Temple architecture **IMPORTANT**

Parts: **Garbhagrha** (sanctum, "womb chamber"), **Śikhara/Vimāna** (tower), Antarāla (vestibule), Maṇḍapa (hall), Prākāra (enclosure), **Gopuram** (gateway tower, South), Pradakṣiṇā-patha (circumambulatory path), Amalaka + kalaśa (crown, North).

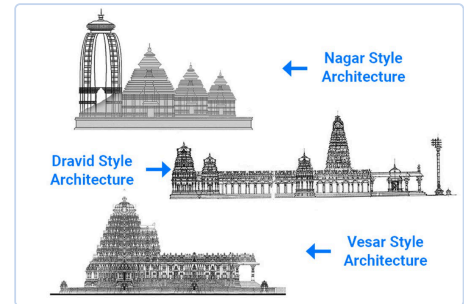
Feature	Nāgara (North)	Drāviḍa (South)	Vesara (Deccan)
Tower	Curvilinear Śikhara + amalaka	Pyramidal Vimāna	Hybrid
Gateway	Small / modest	Massive Gopurams	Moderate
Walls / layout	Early forms no boundary wall; square plan on raised platform (jagatī)	Enclosed complexes, multiple prakāras, temple tank (puṣkariṇī)	Star-shaped plans (Hoysala)
Material	Sandstone	Granite	Soapstone
Examples	Kandariya Mahadeva (Khajuraho), Konark, Lingaraja (Odishan), Modhera (Solanki)	Brihadeeswara (Thanjavur), Meenakshi (Madurai), Shore Temple, Srirangam	Chennakesava (Belur), Hoysaleswara (Halebidu), Aihole, Pattadakal



Nāgara basics

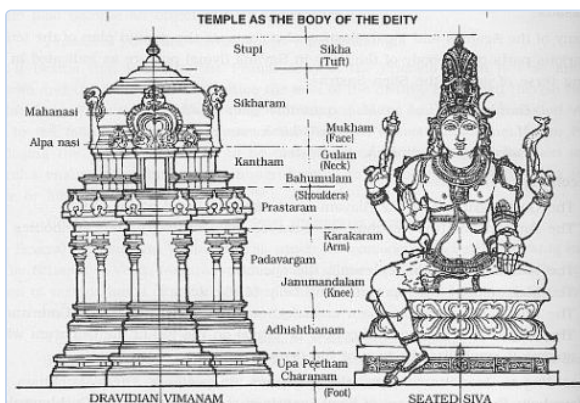


Drāviḍa style

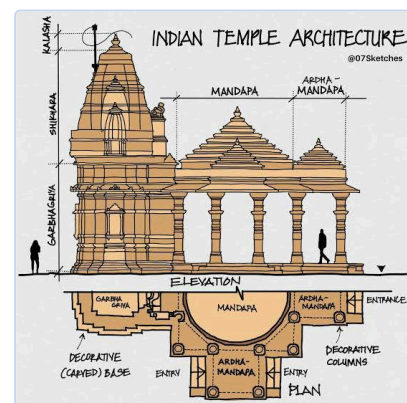


Nāgara · Drāviḍa · Vesara

- **Science in temples (functionality + aesthetics, MQP 8b):** geometry and proportions (tāla-māna, maṇḍala grids; geometry gives stability, harmony and symbolism); astronomy and orientation (sunlight in sanctum on specific days: Konark, Modhera); acoustics (musical pillars at Hampi); material science (mortise-tenon joints, no mortar; polished Chola granite; Hoysala soapstone carving).
- **Symbolism:** temple = macrocosm; sanctum = Brahmāṇḍa/cosmic womb; vertical rise = material to spiritual; circumambulation = cosmic cycles; temple as body of the deity (figure).
- **Applying Vāstu in modern design (SEE 8b):** site and soil analysis, orientation to sun and wind (passive cooling/daylight), functional zoning, natural materials and ventilation, courtyards and water harvesting, human-scale proportions.



Temple as the body of the deity



Parts of an Indian temple

[BOOK p.311 Architecture history ↗](#)

[BOOK p.313 Vāstu-śāstra ↗](#)

[BOOK p.316 Vāstu-puruṣa-maṇḍala ↗](#)

[BOOK p.318 Eight limbs ↗](#)

[BOOK p.321 Town planning ↗](#)

[BOOK p.327 Temple architecture ↗](#)

UNIT 5 · Governance, Public Administration & Sustainability

5.1 Rāmāyaṇa on governance (Rāma-Rājya) **IMPORTANT**

- Vālmīki's epic is also a treatise on leadership and ethical administration. **Rāma-Rājya** = ideal governance: moral authority, citizen welfare (**lokakalyāṇa**), participatory administration. Aligned with transparency, accountability, equity, service. **Servant leadership**: citizens before self.
- Principles**: (i) **citizen-centric administration**; (ii) **advisory councils / consultative governance**: Rāma consulted ministers like Vasiṣṭha (SEE 9a); expert advice → legitimacy and effectiveness; (iii) **ethical law enforcement**: fair punishment/reward (nyāya); (iv) **crisis management** in exile: limited resources, allies, preparation (disaster preparedness, coordination); (v) **sustainability and welfare**: long-term prosperity, ecological balance.
- Case**: after return to Ayodhyā: audits of administration/resources, welfare for all castes and classes, transparent justice, counselling by ministers and citizens → peace, prosperity.

Rāmāyaṇa principle	Modern lesson
Dharma & Nyāya	Ethical policy, legal frameworks
Servant leadership	Officials put citizens before ambition
Team management & delegation	Bureaucratic structure, empowerment
Crisis management	Disaster/contingency planning
Visionary planning	Sustainable development
Communication & diplomacy	Negotiation, stakeholder engagement
Empathy & social justice	Inclusive policies

Key attributes (CIE-3): justice, compassion, dharma, truthfulness, welfare. **Rāja-dharma** = ruler's moral duty to govern justly and protect people. **Lokasaṅgraha** = social welfare / holding the world together (MCQ B). Hindu governance: king's happiness lies in **welfare of subjects**; the king as **servant of the state**.

[BOOK p.377 Rāmāyaṇa attributes ↗](#)

[BOOK p.398 Epics & administration ↗](#)

5.2 Arthaśāstra (Kauṭilya / Cāṇakya, 4th c. BCE) **IMPORTANT**

- First comprehensive manual of public administration: statecraft, economics, law, taxation, military, foreign policy. Blends **dharma** (ethics) with **artha** (pragmatism).
- Eight governance principles**: sovereignty & rule of law · welfare orientation (lokakalyāṇa) · systematic bureaucracy (amātya, treasurers, tax collectors) · strategic/pragmatic leadership · financial administration · intelligence & surveillance (spies) · law & justice · ethics & morality.
- Principles of effective governance**: **Ṣaḍguṇya** (sixfold policy): **sandhi** (peace), **vigraha** (war), **āsana** (neutrality), **yāna** (march), **saṁśraya** (seek protection / alliance), **dvaiddhībhāva** (dual policy); fourfold responsibility of king (protect, secure wealth, justice, dharma); planning and strategy; ethics + pragmatism.
- Saptāṅga theory of the state**: **Svāmin** (king), **Amātya** (minister), **Janapada** (territory & people), **Durga** (fort), **Kośa** (treasury), **Daṇḍa** (law/force), **Mitra** (ally). (CIE-3: "Dharma is not an element".)

Element	Meaning	Modern equivalent
Janapāda	"Land of people": territory + people; basic unit for resources, tax, law. Mahājanapadas (Magadha, Kosala, Vatsa...)	State/district administration
Durga	Fort / fortified capital: defence and admin centre. (Symbolic: Durgā over Mahiṣāsura = order over chaos)	Defence, strategic hubs
Kośa	Treasury: revenue, wealth; "treasury is more important than the army" (Kauṭilya); funds everything	Finance dept., budget
Daṇḍa	"Stick": authority, law enforcement, punishment; balances power and justice	Police, judiciary
Mitra	"Friend": alliances, diplomacy (ṣaḍguṇya)	Foreign policy

Interrelation: Janapada defines the territory → Durga protects it and the Kośa → Kośa funds administration, army, welfare → Daṇḍa maintains order and secures resources → Mitra safeguards the realm and trade. (CIE-3 4b)

IMPORTANT State sources of income per Arthaśāstra (SEE Feb 2026 10b; textbook p.388–389)

- **Seven heads:** city, country, mines, irrigation works, forests, cattle-herds, trade routes.
- **Kośa sources:** **Mūlya** (price of state goods), **Bhāga** (share of produce, 1/6th), **Vyājī** (tax on sales), **Parigha** (protective duty), **Kṛpta** (fixed levy at river-banks/ports), **Rūpika** (surcharge on manufactures), **Atyaya** (penalties/fines).
- Land tax 1/6 of agricultural produce; water-usage tax 1/5–1/3; customs (śulka: import/export), excise, toll, pasture leases, fines; mines and state enterprises; labour (viṣṭi).

[BOOK p.379 Arthaśāstra ↗](#)

[BOOK p.381 Kauṭilyan state ↗](#)

[BOOK p.385 Amātya & Janapada ↗](#)

[BOOK p.386 Durga ↗](#)

[BOOK p.387 Kośa ↗](#)

[BOOK p.390 Daṇḍa ↗](#)

[BOOK p.392 Mitra ↗](#)

[BOOK p.396 Relevance ↗](#)

5.3 IKS and the UN SDGs **IMPORTANT**

- **17 SDGs** adopted by the UN in 2015, target 2030. Shared foundation with IKS: sustainability, equity, intergenerational well-being; **Vasudhaiva Kuṭumbakam** ↔ "leave no one behind". IKS holistic worldview = **Pūrṇatva**.

IKS principle	Essence	SDG
Dharma	Righteous duty, social order	16 Peace, Justice, Institutions
Artha	Sustainable prosperity	8 Decent work & growth
Kāma	Responsible enjoyment	3 Health & well-being
Mokṣa	Inner peace	4 Quality education
Pañca-mahābhūta	Balance with nature	13, 14, 15 Climate, water life, land life
Sarvodaya	Welfare of all	1, 2, 10
Ahiṃsā	Non-violence	16
Swadeshi	Local self-reliance	9, 11, 12

- **Examples:** SDG1: Gram Swaraj. SDG2: Vṛkṣāyurveda, Panchagavya, **Sikkim 100% organic**. SDG3: Ayurveda/Yoga; **International Yoga Day 21 June** (SEE 9b: Ayurveda ↔ SDG 3). SDG4: Gurukula; NEP 2020. SDG5: Gārgī, Maitreyī, Lopāmudrā. SDG6: stepwells, tanks, **Rajasthan johads**. SDG 7/9/11/12: passive cooling in Hawa Mahal/temples. SDG 13–15: sacred groves (Devrai/Kavu). SDG 16/17: Sabhā and Samiti; Mitra.
- **Sustainable governance** = meeting present needs without harming the future (intergenerational equity). Dharma → justice, equality, sustainability.

5.4 Safeguarding traditional knowledge **IMPORTANT**

- **Threats:** **biopiracy**, cultural erosion (languages/oral tradition), lack of documentation, inadequate IPR recognition for community knowledge, global-market exploitation (MNCs).
- **Methods:** (1) digitalisation/documentation (proves "prior art"), (2) legal frameworks/IPR, (3) education integration (NEP), (4) community empowerment.

Mechanism	Role
TKDL (Traditional Knowledge Digital Library)	Codified, searchable database of traditional medicinal knowledge accessible to global patent offices → stops biopiracy / false patents (MQP: to protect Indian TK). Expanding to Yoga, manuscripts, food.
Biological Diversity Act 2002; National Biodiversity Authority	Conservation, benefit sharing; regulates foreign access.
GI Act 1999	Protects origin-specific products (Darjeeling tea, Kanchipuram silk, Mysore sandalwood).
AYUSH, Ministry of Culture, IKS Division (AICTE/MoE), National Mission for Manuscripts, CBBRs	Codification, education, validation, preservation.

IMPORTANT Case studies (SEE 10a: role of TKDL)

- **Turmeric:** US Patent 5,401,504 (1995, Univ. of Mississippi: wound healing). CSIR requested re-examination with Sanskrit/English evidence of prior use → **patent revoked 1997**.
- **Neem:** European Patent 0436257 B1 (1994, W.R. Grace + US Dept of Agriculture: fungicide). Opposed by Indian farmers/activists (RFSTE, Greenpeace) at EPO → **revoked 2000**: lacked novelty and inventive step.
- Lesson: documentation + legal tools can defend TK; TKDL makes proof instantly available to examiners.

Why safeguard now (globalisation): protects community rights and ownership, prevents unauthorised patents, ensures benefit sharing, preserves cultural identity, supports sustainable development.

QUICK REVISION · Numbers, Mnemonics & Rapid-Fire MCQs

Numbers to remember

Fact	Number	Fact	Number
Pāṇini's sūtras	3,983 (8 ch × 4 pādas)	Maheśvara Sūtras	14
Ṛgveda hymns / verses / maṇḍalas	1,028 / 10,552 / 10	Sāmaveda verses	1,549
Atharvaveda hymns / mantras / books	730 / ~6,000 / 20	Vedāṅgas / Darśanas (āstika)	6 / 6
Gāyatrī / Anuṣṭup syllables	24 / 32	Gaṇas (Piṅgala)	8 (3-bit words)
Brahmagupta rules for zero	628 CE	Āryabhaṭīya	499 CE, 121 verses, 4 pādas
Mahāyuga	4,320,000 yr	Pañchāṅga limbs	5 (tithi, vāra, nakṣatra, yoga, karaṇa)
Tithis in a lunar month	30 (12° each)	Nakṣatras / yogas	27 (13°20')
Karaṇas	11 (4 fixed + 7)	Adhika māsa	every ~32–33 months
Wootz carbon	1–1.5%	Delhi Pillar	7.2 m, 99.7% Fe, 5th c. CE
Zinc distillation (Zawar)	9th–13th c., ~1000 °C	Crucible steel furnace	~1400 °C
π values	3.088 · 3.1416 · 3927/1250 · 11 dp (Mādhava)	Sine table (Āryabhaṭa)	24 intervals of 3.75°, R = 3438'
Dhanus / Gṛhapatya dhanus	96 / 108 aṅgula	Harappan brick ratio	4 : 2 : 1
64 Kalās · Pañcakośa	64 · 5	Eight limbs of Vāstu	8 (Sthāpatya → Ābharaṇa)
Saptāṅga	7 elements	Śāḍgunya	6 policies
SDGs	17 (UN, 2015)	Turmeric / Neem patents revoked	1997 / 2000

Mnemonics

- **Vedāṅga body:** Chandas = feet, Kalpa = hands, Jyotiṣa = eyes, Nirukta = ears, Śikṣā = nose, Vyākaraṇa = mouth.
- **Pañcāvayava: P-H-U-U-N:** Pratijñā, Hetu, Udāharaṇa, Upanaya, Nigamana.
- **Fallacies: S-V-P-S-K:** Savyabhicāra, Viruddha, Prakaraṇasama, Sādhyasama, Kālātīta.
- **Pañchāṅga: T-V-N-Y-K** (Tithi, Vāra, Nakṣatra, Yoga, Karaṇa). **Rāśi is NOT one.**
- **Saptāṅga: S-A-J-D-K-D-M:** Svāmin, Amātya, Janapada, Durga, Kośa, Daṇḍa, Mitra.
- **Dosha elements:** Vāta = Air+ether, Pitta = Fire, Kapha = Water+earth.
- **Śāḍgunya:** Sandhi, Vighraha, Āsana, Yāna, Saṃśraya, Dvaidhībhāva.
- **Guru conditions:** Long, Cluster, Anusvāra/visarga, Last.

Rapid-fire (likely Part A)

Question	Answer
IKS means (SEE 1.2)	Indigenous systems of knowledge evolved in India over millennia
Pratyakṣa (SEE 1.3)	Direct perception
Western education in India emphasised	Discipline-centric (specialised) knowledge, replacing integrated learning
Takṣaśilā and Nālandā	Ancient centres of higher learning
Which text is in the Vedic corpus?	Upaniṣads (Purāṇas, Smṛti, Itihāsa are not)
Sā vidyā yā vimuktaye / Satyaṃ vada dharmaṃ cara	Knowledge leads to liberation / Taittirīya Upaniṣad
Indian philosophical systems differ because they	Encourage multiple viewpoints (and debate)
Language of knowledge transmission; traditional education centre	Sanskrit; Gurukula system
Primary mode of Gurukula learning	Debate and memorisation (Śravaṇa, Manana, Nididhyāsana)
Piṅgala's binary ideas came from studying	Chandas (metre)
Primary character of Śruti	Oral tradition / revelation
Vedāṅga for phonetics	Śikṣā
Vedāṅga called "mouth"	Vyākaraṇa
Philosophy of logic & epistemology	Nyāya
Component dealing with philosophical inquiry into Ātman	Upaniṣads
Transition trad. → Western knowledge (colonial)	Epistemological shift
Ṭarka (as in philosophical systems)	Dialectical reasoning
TK vs modern science	TK holistic & community-based; modern reductionist
Veda that is "Book of Songs"	Sāmaveda (1,549 verses)
Śukla vs Kṛṣṇa Yajurveda	Mantra & brāhmaṇa separate vs mixed
Cārvāka accepts only	Pratyakṣa
Anekāntavāda	Jainism
Author of Aṣṭādhyāyī / sūtras	Pāṇini / 3,983 (not 3000, 3500, 4200)
Maheśvara Sūtras purpose	Organise phonemes into groups
Style of Pāṇini's grammar	Sūtra style
Suffixes for nouns / verbs	sup / tiṅ
Basic element for word generation	Dhātu / prātipadika
Sthāna	Place of articulation
Bhūta-saṃkhyā uses	Objects, beings, concepts
Kaṭapayādi	Consonants encode numbers; memory aid (not encryption)
Pramāṇa by similarity	Upamāna
Which is a pramāṇa? (Prameya/Saṃśaya/Pratyakṣa/Saṃskāra)	Pratyakṣa
Steps in Anumāna	Five
Objects of valid knowledge	Prameya
Process of inquiry starts with	Saṃśaya (doubt)
Pingala's gaṇas ↔ modern	de Bruijn sequence

Laghu : Guru in binary	1 : 0
Meru-prastāra ↔	Pascal's triangle
Zero as placeholder → number thanks to	Place-value system
Rules for zero & negative numbers	Brahmagupta
Mathematics compared to	Crest on peacock's head
Śulba Sūtras relate to	Geometry
Sine in ancient India	Jyā
Earth rotates on axis	Āryabhaṭa
Primary source of Vedic astronomy	Vedāṅga Jyotiṣa
Instrument for shadows / time	Gnomon (śaṅku) / Ghaṭī yantra (water clock)
Nakṣatras in the sky	27 lunar mansions
Wootz steel	High-carbon; Damascus swords
Lost-wax casting	Cire perdue; wax = beeswax
Sand-bath yantra	Vāluka yantra
Indigo from	Leaf
Perfume science	Gandhayukti / Gandhaśāstra; "attar" (itr)
Vāta / Pitta / Kapha elements	Air+ether / Fire+water / Water+earth
Dominant age: Kapha / Pitta / Vāta	Childhood / Adulthood / Old age
Kapha imbalance guṇa	Tamas
Meditation & ethical living raise	Sattva
BMI-C information flow	Senses → Mind → Intellect → Action
Sleep & nutrition improve first	Body
Most famous 64-Kalā list	Kāma Sūtra (Vātsyāyana)
Vāstu treatise / basic unit	Mānasāra / Pada
Best direction in Vāstu	East
Harappan planning	Grid + drainage
Garbhagrha / Vimāna	Sanctum / sanctum tower
Gopurams	Drāviḍa (gateway towers)
Curvilinear śikhara · star plan	Nāgara · Hoysala (Vesara)
Water management of South India	Tanks (ēri)
Rāmāyaṇa governance	Welfare-oriented
Janapada / Durga / Kośa / Daṇḍa / Mitra	Territory+people / fort / treasury / punishment-law / ally
Not in Saptāṅga	Dharma
Internal security element	Daṇḍa
Arthaśāstra views king as	Servant of the state
Text of diplomacy, war, peace	Arthaśāstra
TKDL	Protect Indian traditional knowledge (against biopiracy)
IKS aligned with SDGs via	Sustainability; Sarvodaya
Ayurveda ↔ which SDG	SDG 3 Good Health & Well-being

PART B · Answer Outlines for Likely 8-Mark Questions

Each (a) and (b) is 4 marks: write 6–8 crisp points, one diagram or table if relevant. Order below follows unit order.

Question	Outline (what to write)
U1 Vedic corpus; Vedāṅgas as supporting sciences (MQP 1a)	Define Veda (vid), Śruti, 4 Vedas with features → 4 divisions → 6 Vedāṅgas with body analogy and one line each → conclude: Vedāṅgas help in recitation, interpretation, ritual (Śikṣā-Chandas-Vyākaraṇa-Nirukta-Kalpa-Jyotiṣa).
U1 Holistic IKS vs reductionist West (1b)	Table: holistic/experiential/community/oral/sustainable vs analytic/empirical/reductionist/documented/control of nature; examples Ayurveda vs allopathy; need for synthesis.
U1 Historicity of IKS (SEE 1b)	Vedic period → epics/classical → philosophy → maths/astronomy/medicine → Nalanda/Takshashila → manuscripts → colonial disruption → NEP 2020 revival.
U1 TKDL and biopiracy (MQP 2b)	Define biopiracy → turmeric/neem cases → TKDL (database, prior art, patent offices) → outcome.
U2 Pāṇini as computational grammar (MQP 3a / SEE 3a)	Aṣṭādhyāyī facts (3,983, 8×4) → root + suffix + rule → recursion/anuvṛtti → CFG/BNF table → example Rāmaḥ phalam khādati → generative power (new words) → NLP/AI relevance.
U2 Piṅgala & binary / de Bruijn (MQP 3b / SEE 4b)	Laghu/Guru and 4 guru types → L=1/G=0 → 8 gaṇas table → mnemonic and binary cycle → de Bruijn → prastāra, naṣṭa, uddiṣṭa, meru-prastāra, Pascal.
U2 Knowledge triangle (MQP 4a)	Pramāṭṛ-Prameya-Pramāṇa with figure → Vaiśeṣika (6 padārthas) and Nyāya (4 pramāṇas) → Saṃśaya starts inquiry → tarka → siddhānta → nirṇaya.
U2 Origin of zero (MQP 4b); Decimal place value (SEE 3b)	Evidence timeline: Piṅgala śūnya → Bakhshālī dot → Āryabhaṭa place value → Brahmagupta 628 CE rules → inscriptions 594/876 CE → Arabs (Al-Khwārizmī) → Europe (Fibonacci). Advantages vs Roman numerals.
U2 Anumāna with 5 steps (SEE 4a)	Define anumāna (hetu, vyāpti, pakṣa, sādhyā) → 5-step table with the ball or smoke/fire example.
U3 Śulba Sūtras: ritual → geometry (MQP 5a)	Altar construction needs exact shapes → rope constructions → Baudhāyana theorem → triples, $\sqrt{2}$, π → shape transformations → shift from ritual to reasoning.
U3 Āryabhaṭa's astronomy (MQP 5b)	Āryabhaṭīya structure → rotation of Earth (boat analogy) → eclipses by shadow, Moon's reflected light → sine, π → manda/śighra → influence on later astronomers/Arabs.
U3 Pañchāṅga (MQP 6a / SEE 6a)	Five limbs table → basis (Sun–Moon motion, 12°, 27 nakṣatras) → adhika māsa → uses (muhūrta, festivals, farming) → science + culture.
U3 Wootz (MQP 6b); Delhi Pillar (SEE 6b)	See starred boxes: process, properties, Damascus, decline; pillar facts and passive layer.
U3 Four mathematicians (SEE 5a); perfumes (SEE 5b)	Choose Āryabhaṭa, Brahmagupta, Bhāskara II, Mādhava/Piṅgala with work and contribution; perfume: sources, deg-bhapka, fixatives, uses, trade.
U4 Tridoṣa physical + psychological (MQP 7a); Pitta (SEE 7a)	Elements and functions table → prakṛti/vikṛti → aggravation signs → link to guṇas (psychological) → balancing.
U4 64 Kalās (MQP 7b / SEE 8a)	Meaning, source (Kāma Sūtra), categories table → social/economic life (skills, crafts, trade) → modern holistic education.
U4 Irrigation as sustainable water mgmt (MQP 8a)	Harappan reservoirs → Ahar-Pyne → tanks, Kallanai → stepwells, johads → community upkeep → modern relevance.
U4 Architecture: functionality + aesthetics (MQP 8b); Nāgara vs Drāviḍa (SEE 7b)	Comparison table + examples; geometry, acoustics, astronomy, materials; symbolism. Vāstu in modern design (SEE 8b).
U5 Safeguarding TK in globalisation (MQP 9a)	Threats → methods → TKDL, GI, BDA → turmeric/neem → importance.
U5 Dharma-based governance of Rāmāyaṇa (MQP 9b); advisors (SEE 9a)	Principles (citizen-centric, councils, ethical law, crisis, sustainability) + modern table; Vasiṣṭha and ministers.

U5 Traditional governance → sustainable development (MQP 10a/b)

Saptāṅga, Kośa, lokakalyāṇa, IKS–SDG table, integration with modern frameworks (policy, education, local self-reliance).

Last checks before you sleep

- Conventions: **Laghu = 1, Guru = 0**. Digits in saṃkhyā systems are read **right to left**. Sūtras = **3,983**. Pañchāṅga ≠ Rāśi.
- Conflicts flagged above (Vāstu limbs, π value, Piṅgala's date, khādri-rāmāgnayaḥ) are exactly where marks are lost: keep the stated answer.
- If a textbook link does not jump, open the textbook PDF yourself and go to PDF page = printed page $\div 2 + 17$ (rounded down).

All the best. Prepared from your own course material; a few textbook-sourced extras (irrigation, Kośa revenue, Prameya detail, Piṅgala algorithms) are marked so you know where they come from.